

Additional Problems on Linear Equations and Their Graphs

For each problem below, write an equation of the line with the given characteristics.

- ① X-intercept: -5 y-intercept: 2 ② X-intercept: 1 y-intercept: 3 ③ X-intercept: -2 y-intercept: -4

$(-5, 0) \text{ \& } (0, 2)$

$y = mx + b$

$0 = m(-5) + 2$

$\frac{2}{5} = m$

$y = mx + b$

$y = \frac{2}{5}x + 2$

$(1, 0) \text{ \& } (0, 3)$

$m = \frac{y_2 - y_1}{x_2 - x_1}$

$m = \frac{3 - 0}{0 - 1}$

$m = -3$

$y = mx + b$

$y = -3x + 3$

$(-2, 0) \text{ \& } (0, -4)$

$y = mx + b$

$0 = m(-2) - 4$

$-2 = m$

$y = mx + b$

$y = -2x - 4$

- ④ X-intercept: 6 y-intercept: 1

$(6, 0) \text{ \& } (0, 1)$

$y = mx + b$

$0 = m(6) + 1$

$-\frac{1}{6} = m$

$y = mx + b$

$y = -\frac{1}{6}x + 1$

- ⑤ X-intercept: 8 y-intercept: -2

$(8, 0) \text{ \& } (0, -2)$

$y = mx + b$

$0 = m(8) - 2$

$\frac{2}{8} = m$

$\frac{1}{4} = m$

$y = mx + b$

$y = \frac{1}{4}x - 2$

- ⑥ X-intercept: -1 y-intercept: -1

$(-1, 0) \text{ \& } (0, -1)$

$y = mx + b$

$0 = m(-1) - 1$

$-1 = m$

$y = mx + b$

$y = -x - 1$

- ⑦ X-intercept: 3 Slope: -2

$(3, 0), m = -2$

$y - y_p = m(x - x_p)$

$y - 0 = -2(x - 3)$

- ⑧ Point: $(-1, 2)$ y-intercept: -3

$y = mx + b$

$2 = m(-1) - 3$

$-5 = m$

$y = mx + b$

$y = -5x - 3$

- ⑨ Slope: $-\frac{2}{3}$ X-intercept: 1

$(1, 0), m = -\frac{2}{3}$

$y - y_p = m(x - x_p)$

$y - 0 = -\frac{2}{3}(x - 1)$

- ⑩ Slope: $\frac{1}{2}$ Point: $(7, 1)$

$y - y_p = m(x - x_p)$

$y - 1 = \frac{1}{2}(x - 7)$

- ⑪ y-intercept: $\frac{5}{4}$ Slope: -3

$y = mx + b$

$y = -3x + \frac{5}{4}$

- ⑫ X-intercept: 4 Slope: 5

$(4, 0), m = 5$

$y - y_p = m(x - x_p)$

$y - 0 = 5(x - 4)$

- ⑬ Y-intercept -4 and Parallel to $y = -\frac{6}{5}x + 3$.

The slopes of parallel lines are equal.

$m = -\frac{6}{5} \quad b = -4$

$y = mx + b$

$y = -\frac{6}{5}x - 4$

- ⑭ Y-intercept $\frac{3}{7}$ and Perpendicular to $y = \frac{4}{5}x + 2$.

The slopes of perpendicular lines are negative reciprocals.

$m = -\frac{5}{4} \quad b = \frac{3}{7}$

$y = -\frac{5}{4}x + \frac{3}{7}$

⑮ Y-intercept 1 and parallel to $y = 3x - \frac{1}{2}$.

$$m = 3, b = 1$$

$$y = mx + b$$

$$\boxed{y = 3x + 1}$$

⑰ Y-intercept zero and parallel to $y = \frac{3}{8}x - 10$.

$$m = \frac{3}{8}, b = 0$$

$$\boxed{y = \frac{3}{8}x}$$

⑲ Y-intercept $\frac{3}{4}$ and perpendicular to $y = 3x - 1$.

$$m = -\frac{1}{3}, b = \frac{3}{4}$$

$$\boxed{y = -\frac{1}{3}x + \frac{3}{4}}$$

• ⑲ Parallel to $y = 3x + 5$ and containing the point $(1, 4)$.

$$m = 3, (1, 4)$$

$$y - y_p = m(x - x_p)$$

$$\boxed{y - 4 = 3(x - 1)}$$

⑯ Y-intercept $-\frac{10}{11}$ and perpendicular to $y = -\frac{7}{9}x - 5$.

$$m = \frac{9}{7}, b = -\frac{10}{11}$$

$$y = mx + b$$

$$\boxed{y = \frac{9}{7}x - \frac{10}{11}}$$

⑱ Y-intercept 13 and perpendicular to $y = \frac{5}{7}x + 9$.

$$m = -\frac{7}{5}, b = 13$$

$$\boxed{y = -\frac{7}{5}x + 13}$$

⑳ Y-intercept 2 and perpendicular to $y = -5x$.

$$m = \frac{1}{5}, b = 2$$

$$\boxed{y = \frac{1}{5}x + 2}$$

• ⑳ Perpendicular to $y = -\frac{3}{2}x + 5$ and containing the point $(-3, 2)$.

$$m = \frac{2}{3}, (-3, 2)$$

$$y - y_p = m(x - x_p)$$

$$\boxed{y - 2 = \frac{2}{3}(x + 3)}$$

⑳ Parallel to $y = \frac{7}{4}x - 1$ and containing the point $(-2, 5)$.

$$m = \frac{7}{4}, (-2, 5)$$

$$y - y_p = m(x - x_p)$$

$$\boxed{y - 5 = \frac{7}{4}(x + 2)}$$

㉑ Perpendicular to $y = 2x - 3$ and containing the point $(-2, -2)$.

$$m = -\frac{1}{2}, (-2, -2)$$

$$y - y_p = m(x - x_p)$$

$$\boxed{y + 2 = -\frac{1}{2}(x + 2)}$$

㉒ Parallel to $y = -\frac{2}{5}x - 2$ and containing the point $(-2, -5)$.

$$m = -\frac{2}{5}, (-2, -5)$$

$$\boxed{y + 5 = -\frac{2}{5}(x + 2)}$$

㉓ Perpendicular to $y = x + 1$ and containing the point $(2, 6)$.

$$m = -1, (2, 6)$$

$$\boxed{y - 6 = -(x - 2)}$$

• ㉔ Parallel to $y - 2 = 3(x + 4)$ and containing the point $(7, 3)$.

$$m = 3, (7, 3)$$

$$\boxed{y - 3 = 3(x - 7)}$$

㉕ Perpendicular to $-\frac{7}{8}(x + 1) = y + 3$ and containing the point $(-4, 1)$.

$$m = \frac{8}{7}, (-4, 1)$$

$$\boxed{y - 1 = \frac{8}{7}(x + 4)}$$

② Parallel to $\frac{3}{4}(x-1)=y-5$ and containing the point $(-1, -1)$.

$$m = \frac{3}{4}, (-1, -1)$$

$$y + 1 = \frac{3}{4}(x + 1)$$

③ Perpendicular to $y + 6 = 2(x - 2)$ and containing the point $(-2, 0)$.

$$m = -\frac{1}{2}, (-2, 0)$$

$$y - 0 = -\frac{1}{2}(x + 2)$$

③ Y-intercept 7 and parallel to $-10x + 5y = 8$.

$$5y = 10x + 8$$

$$y = 2x + \frac{8}{5}$$

$$m = 2, b = 7$$

$$y = mx + b$$

$$y = 2x + 7$$

④ Y-intercept zero and perpendicular to $4x - 3y = 6$.

$$-3y = -4x + 6$$

$$y = \frac{4}{3}x - 2$$

$$m = -\frac{3}{4}, b = 0$$

$$y = mx + b$$

$$y = -\frac{3}{4}x$$

③ Y-intercept -2 and parallel to $\frac{3}{5}x + y = -4$.

$$y = -\frac{3}{5}x - 4$$

$$m = -\frac{3}{5}, b = -2$$

$$y = mx + b$$

$$y = -\frac{3}{5}x - 2$$

③ Y-intercept $\frac{4}{5}$ and perpendicular to $8x - 2y = 5$.

$$-2y = -8x + 5$$

$$y = 4x - \frac{5}{2}$$

$$m = -\frac{1}{4}, b = \frac{4}{5}$$

$$y = mx + b$$

$$y = -\frac{1}{4}x + \frac{4}{5}$$

③ Parallel to $7x + 5y = -3$ and containing the point $(-2, 9)$.

$$5y = -7x - 3$$

$$y = -\frac{7}{5}x - \frac{3}{5}$$

$$m = -\frac{7}{5}, (-2, 9)$$

$$y - 9 = -\frac{7}{5}(x + 2)$$

③ Perpendicular to $-2x + 10y = -1$ and containing the point $(-2, -2)$.

$$10y = 2x - 1$$

$$y = \frac{2x}{10} - \frac{1}{10}$$

$$y = \frac{x}{5} - \frac{1}{10}$$

$$y = \frac{1}{5}x - \frac{1}{10}$$

$$m = -5, (-2, -2)$$

$$y + 2 = -5(x + 2)$$

③⑦ Parallel to $-5x+2y=7$ and ③⑧ Perpendicular to $2x-y=2$
containing the point $(1,8)$. and containing the point $(4,-3)$.

$$2y = 5x + 7$$

$$y = \frac{5}{2}x + \frac{7}{2}$$

$$m = \frac{5}{2}, (1,8)$$

$$\boxed{y-8 = \frac{5}{2}(x-1)}$$

$$-y = -2x + 2$$

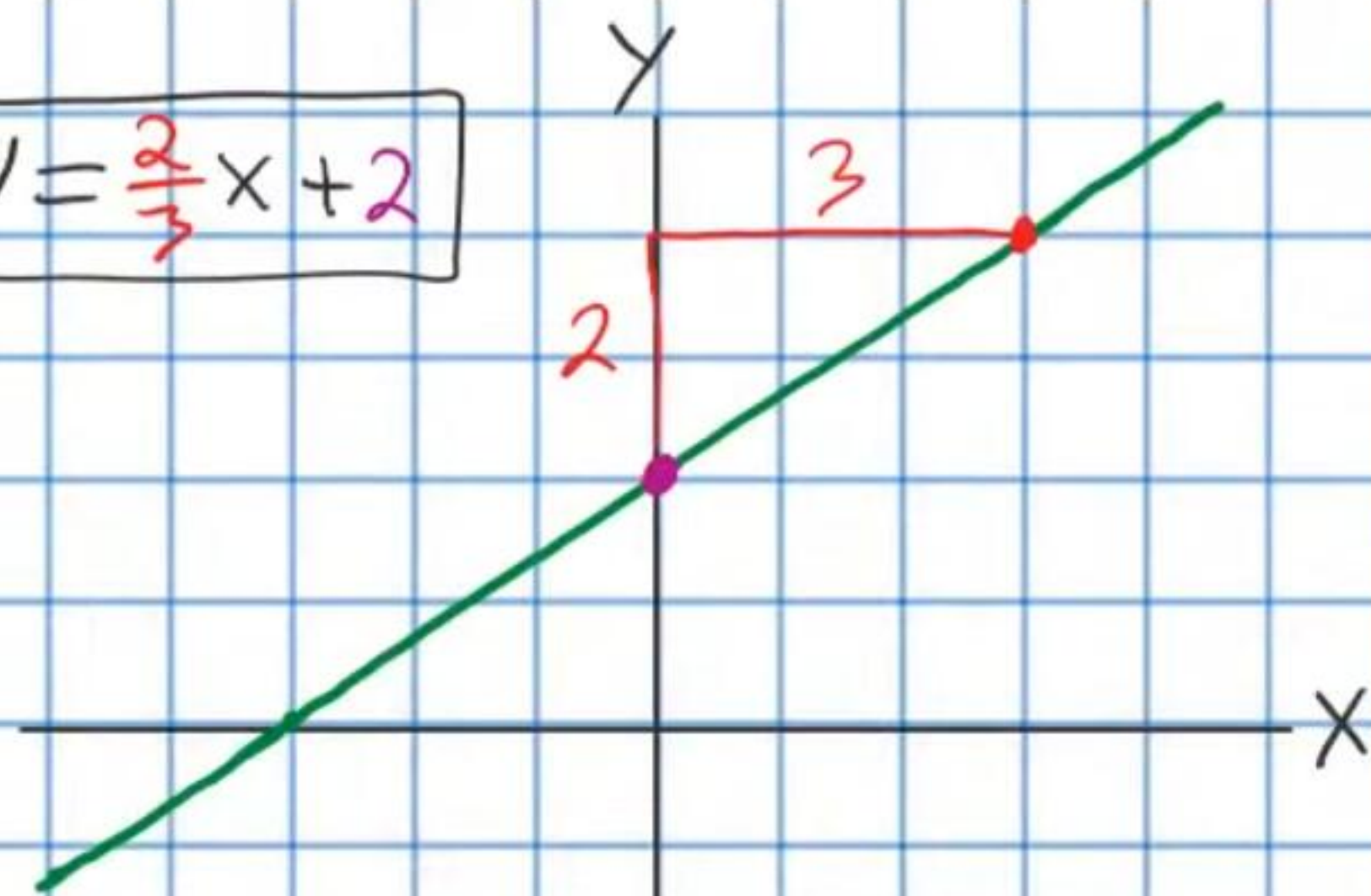
$$y = 2x - 2$$

$$m = -\frac{1}{2}, (4,-3)$$

$$\boxed{y+3 = -\frac{1}{2}(x-4)}$$

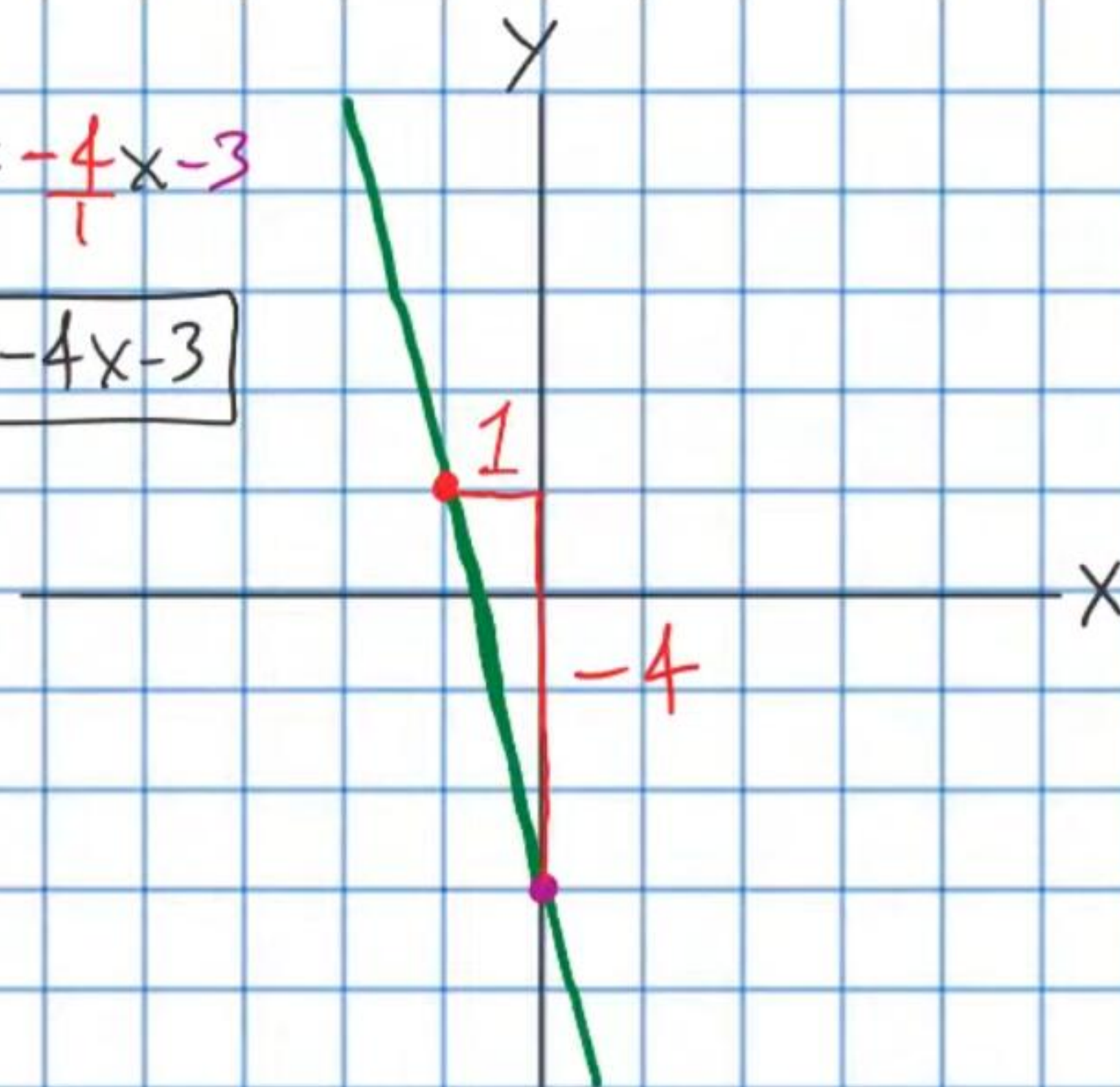
For each graph below, write an equation of the line shown.

③⑨ $y = \frac{2}{3}x + 2$

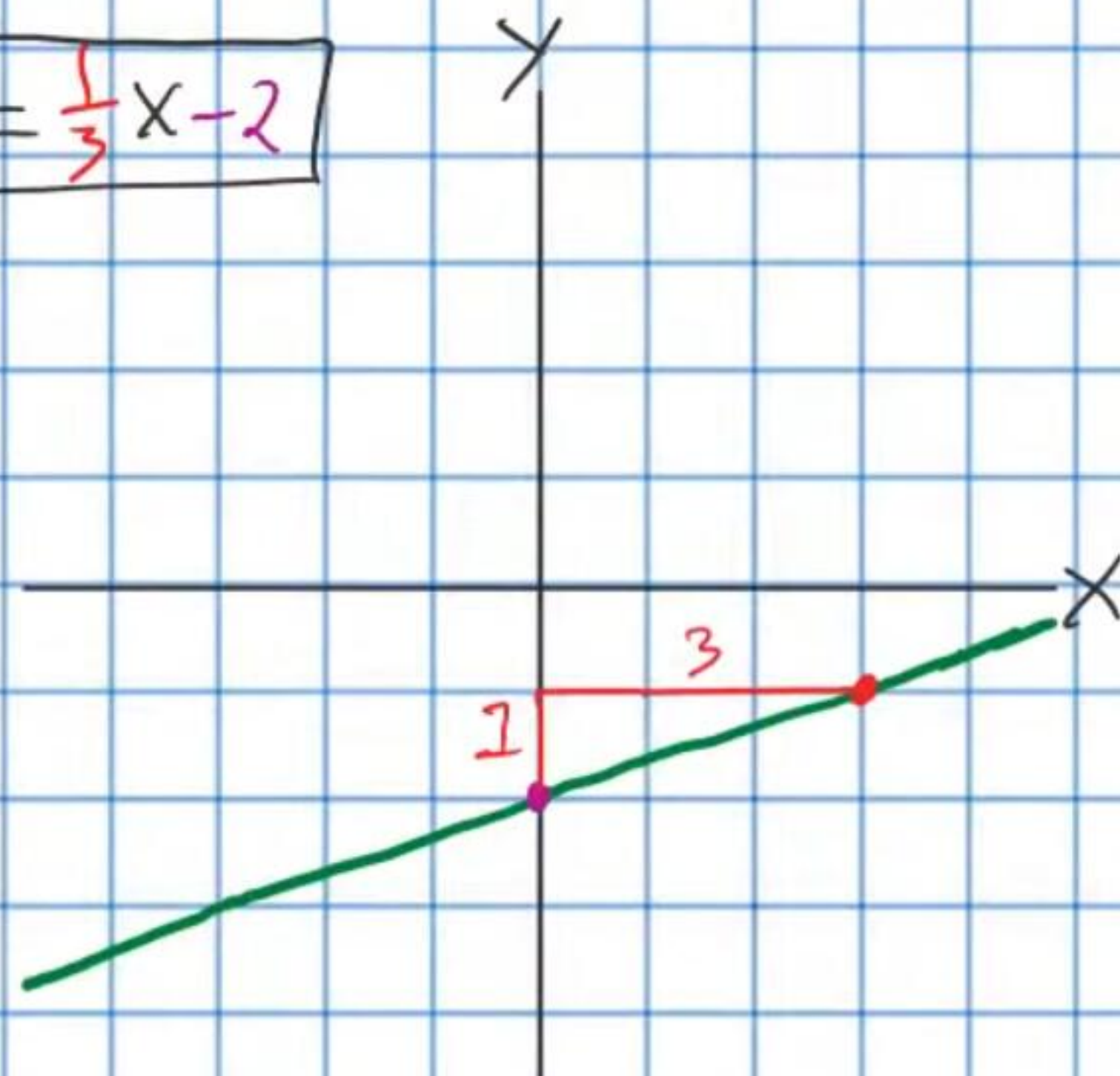


④⑩ $y = -\frac{4}{1}x - 3$

$y = -4x - 3$

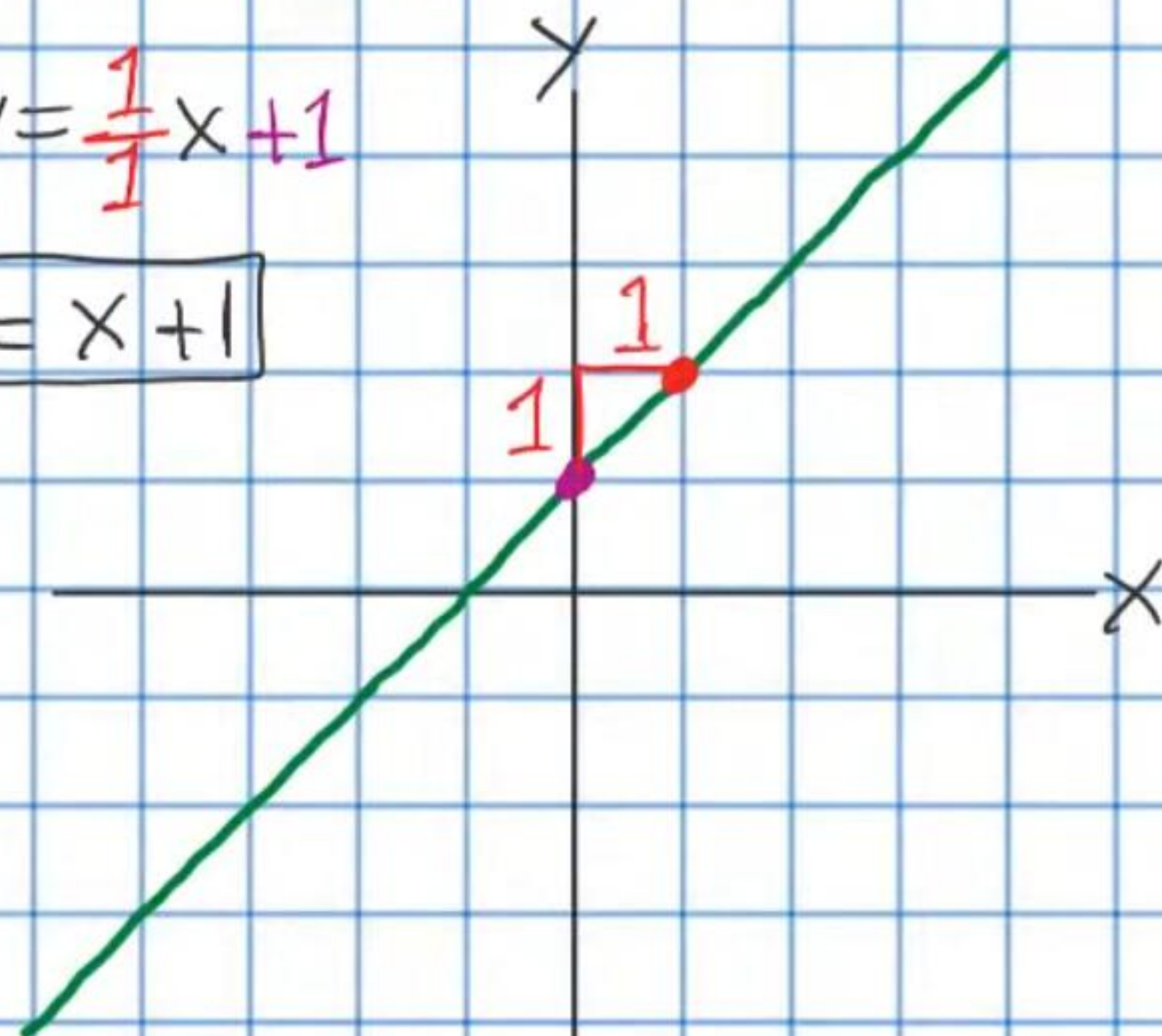


④⑪ $y = \frac{1}{3}x - 2$

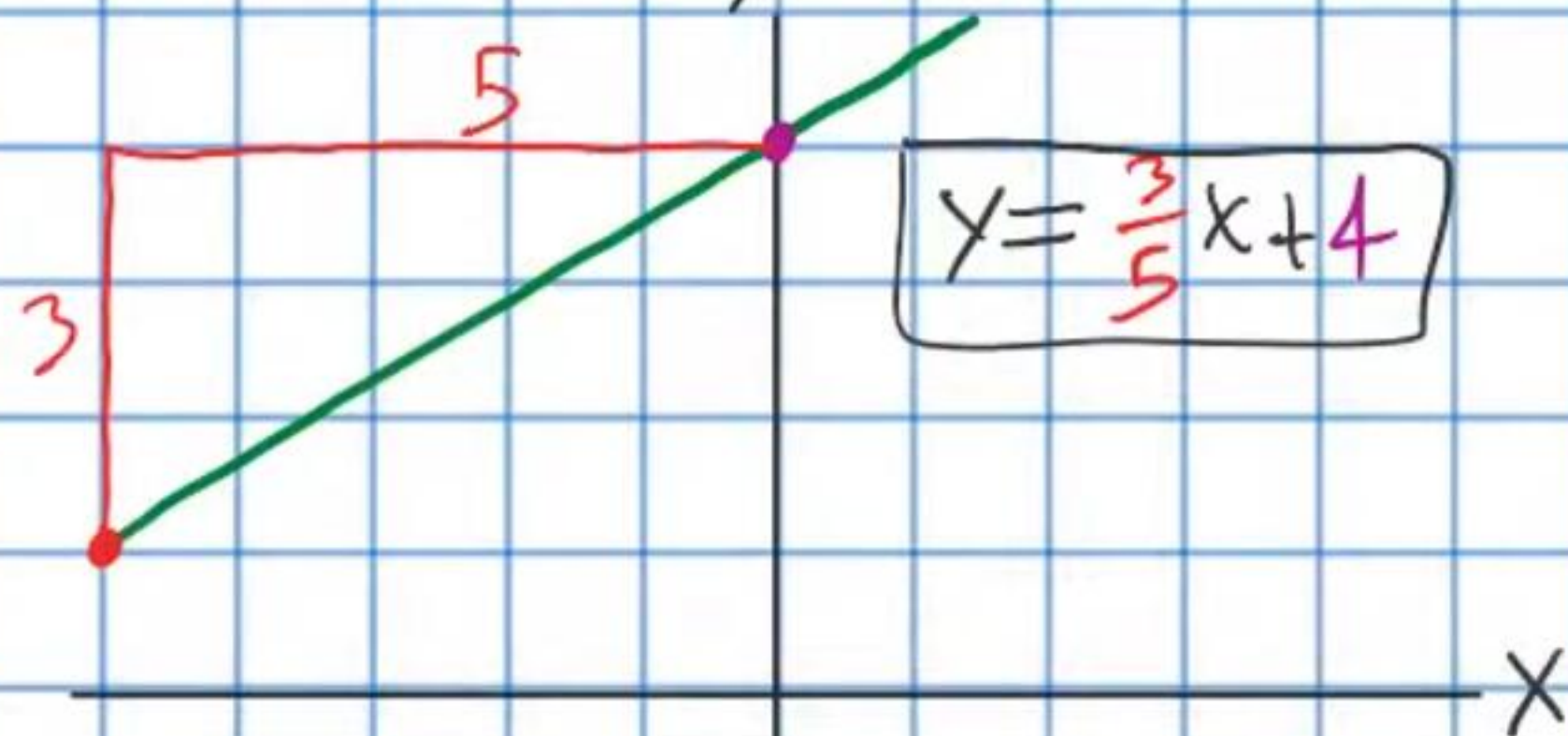


④⑫ $y = \frac{1}{1}x + 1$

$y = x + 1$

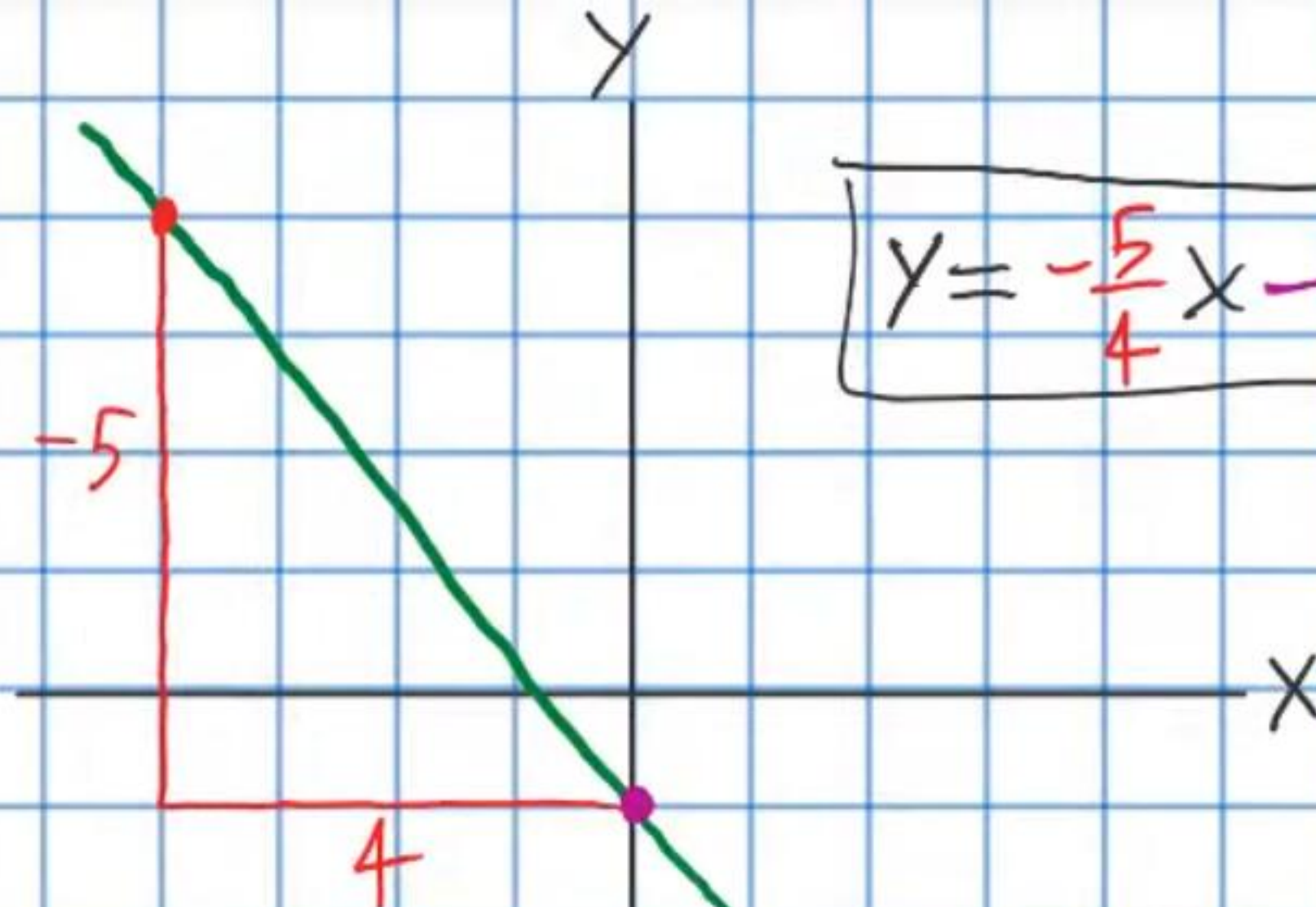


(43)



$$y = \frac{3}{5}x + 4$$

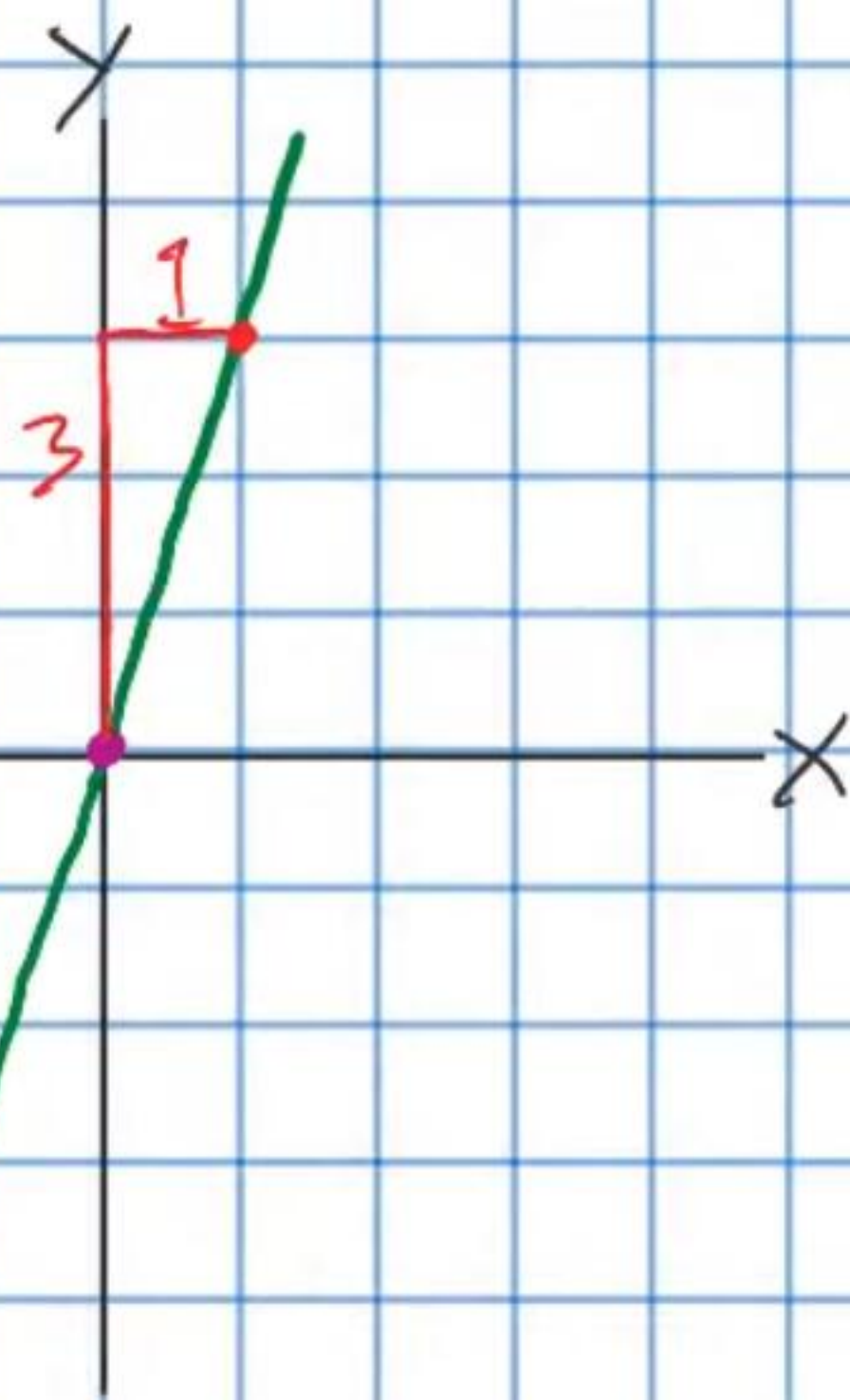
(44)



$$y = -\frac{5}{4}x - 1$$

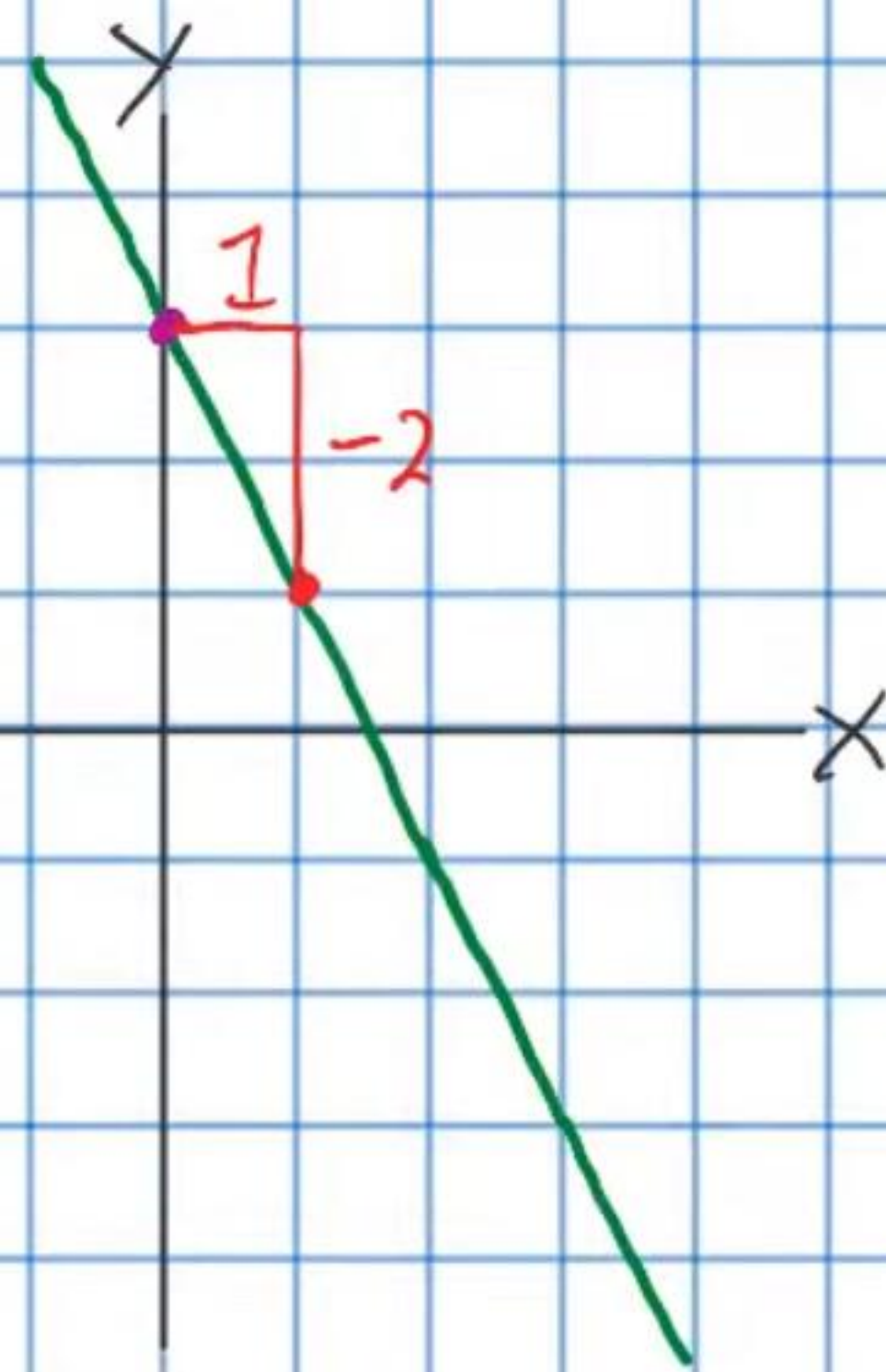
(45) $y = \frac{3}{1}x + 0$

$$y = 3x$$

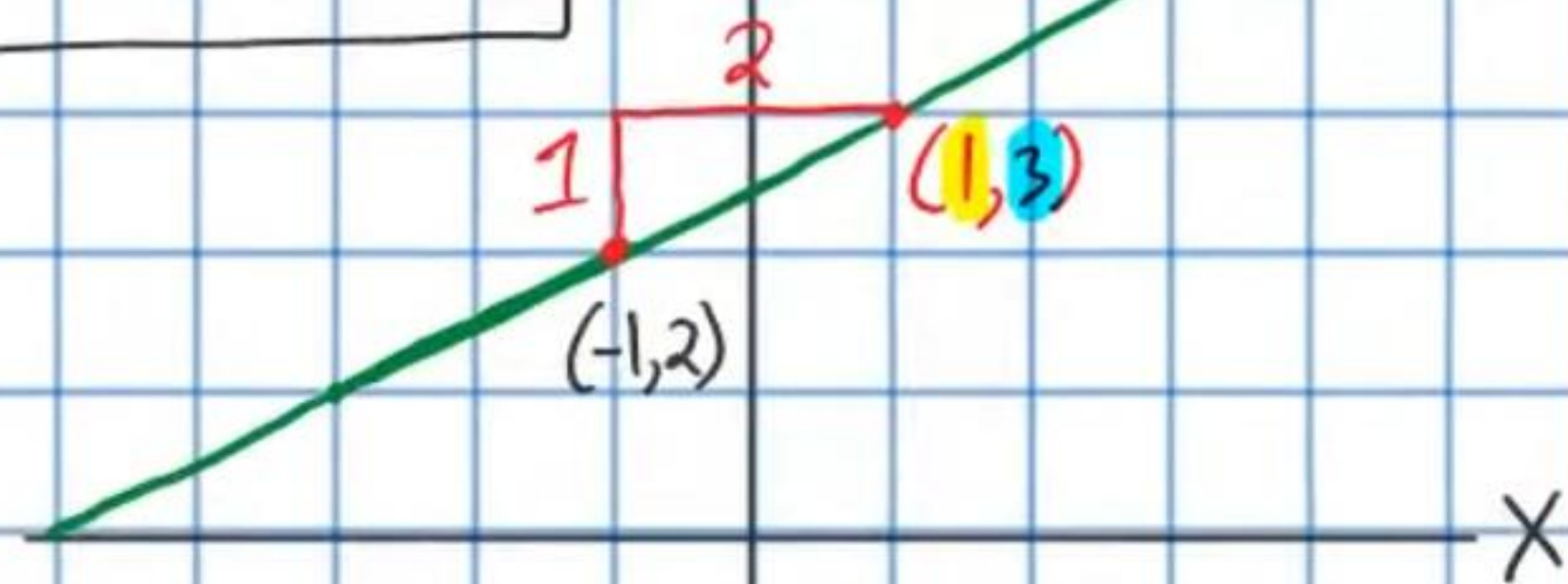


(46) $y = -\frac{2}{1}x + 3$

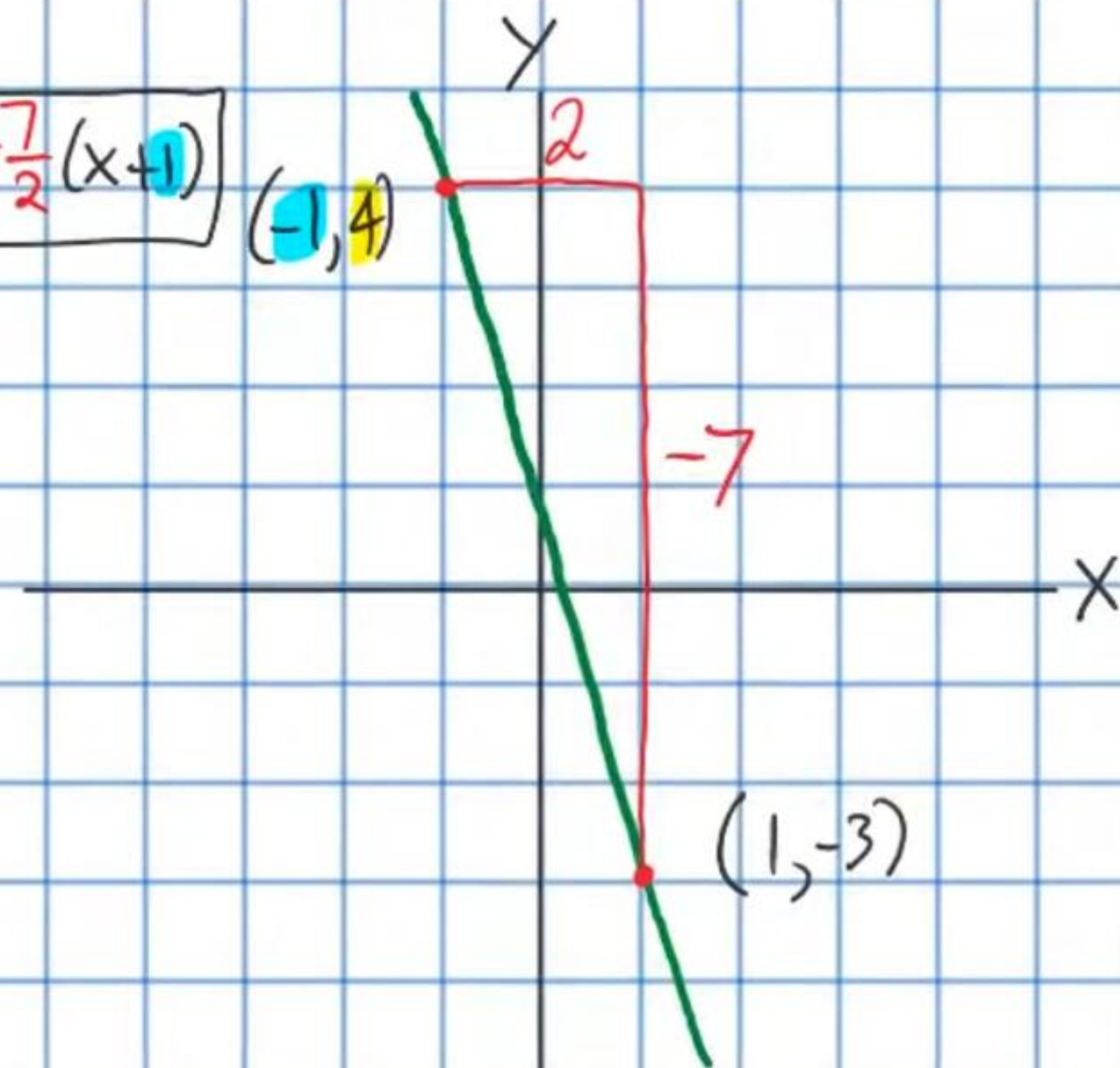
$$y = -2x + 3$$



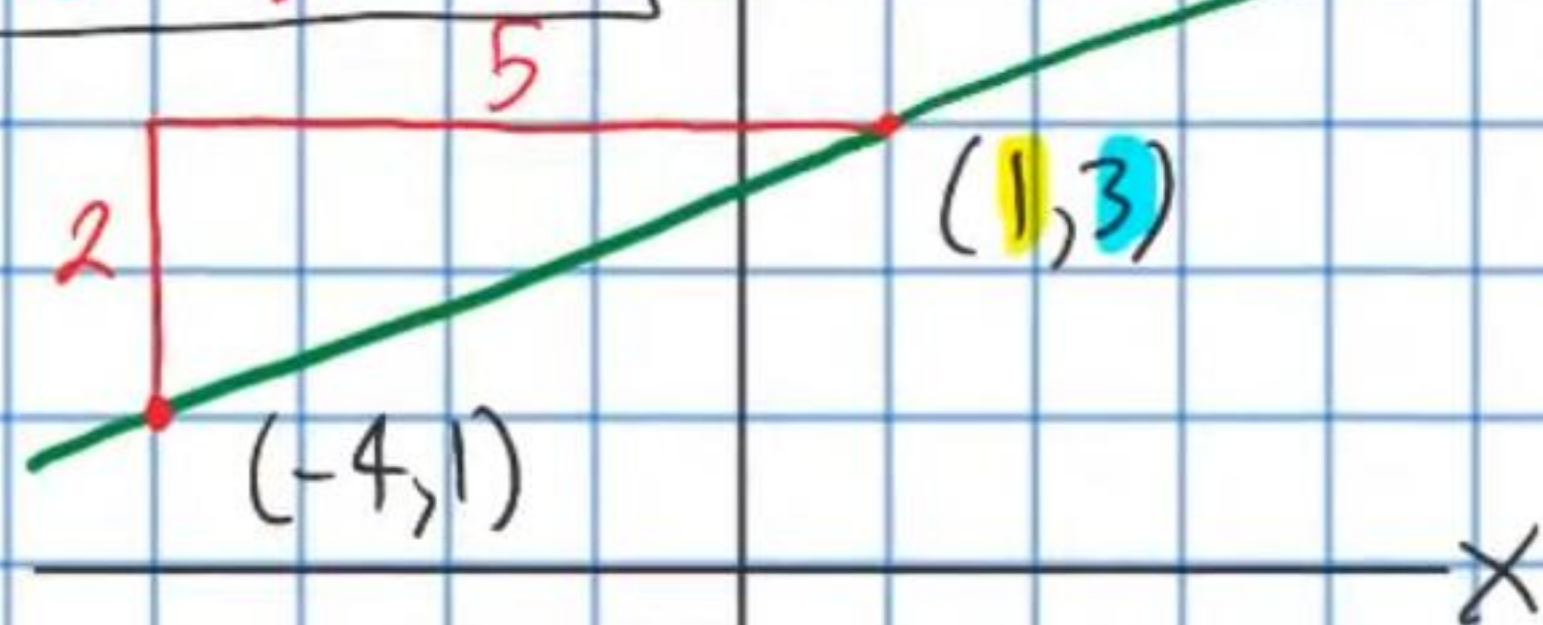
$$(47) \quad y - 3 = \frac{1}{2}(x - 1)$$



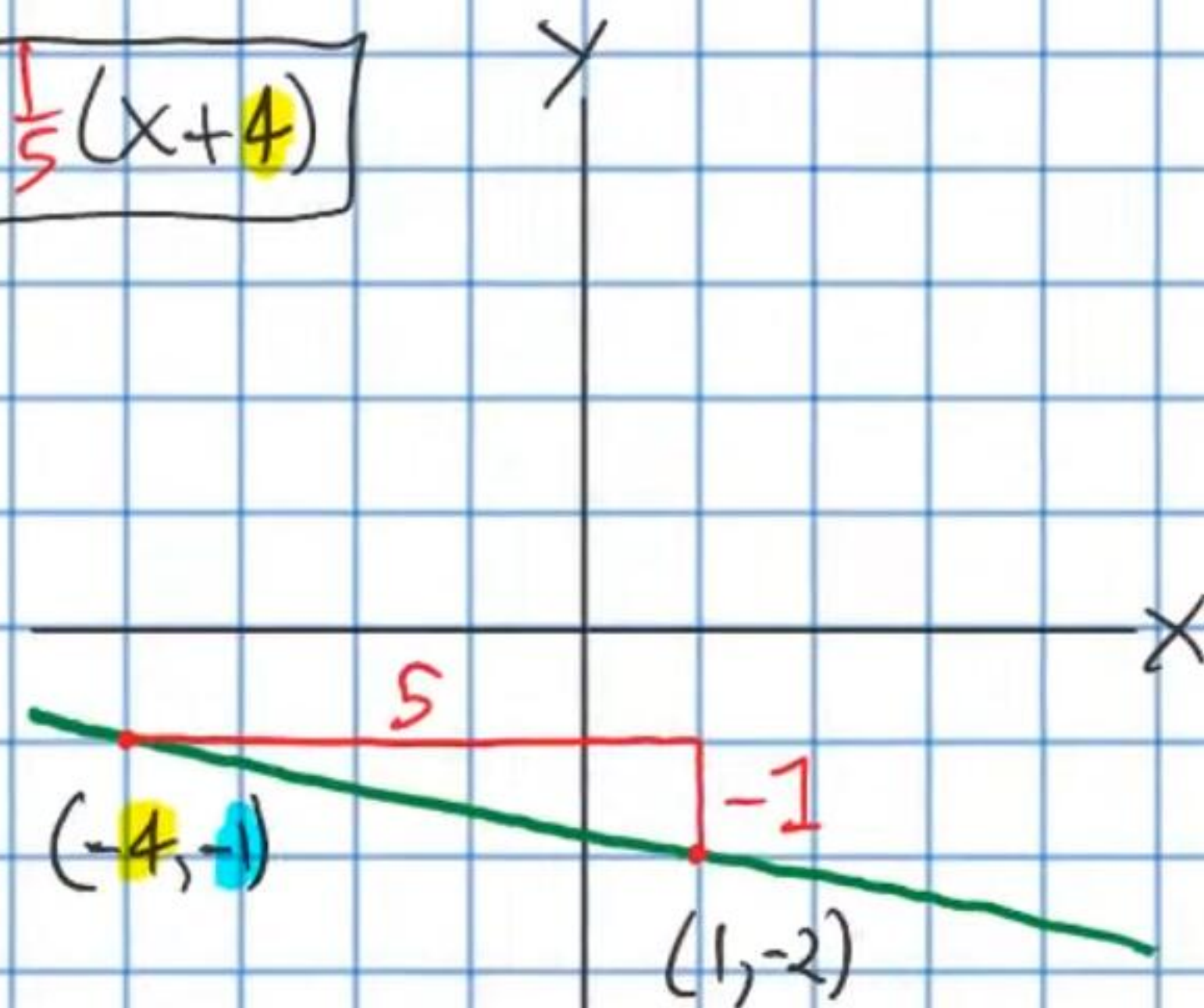
$$(48) \quad y - 4 = -\frac{7}{2}(x + 1)$$

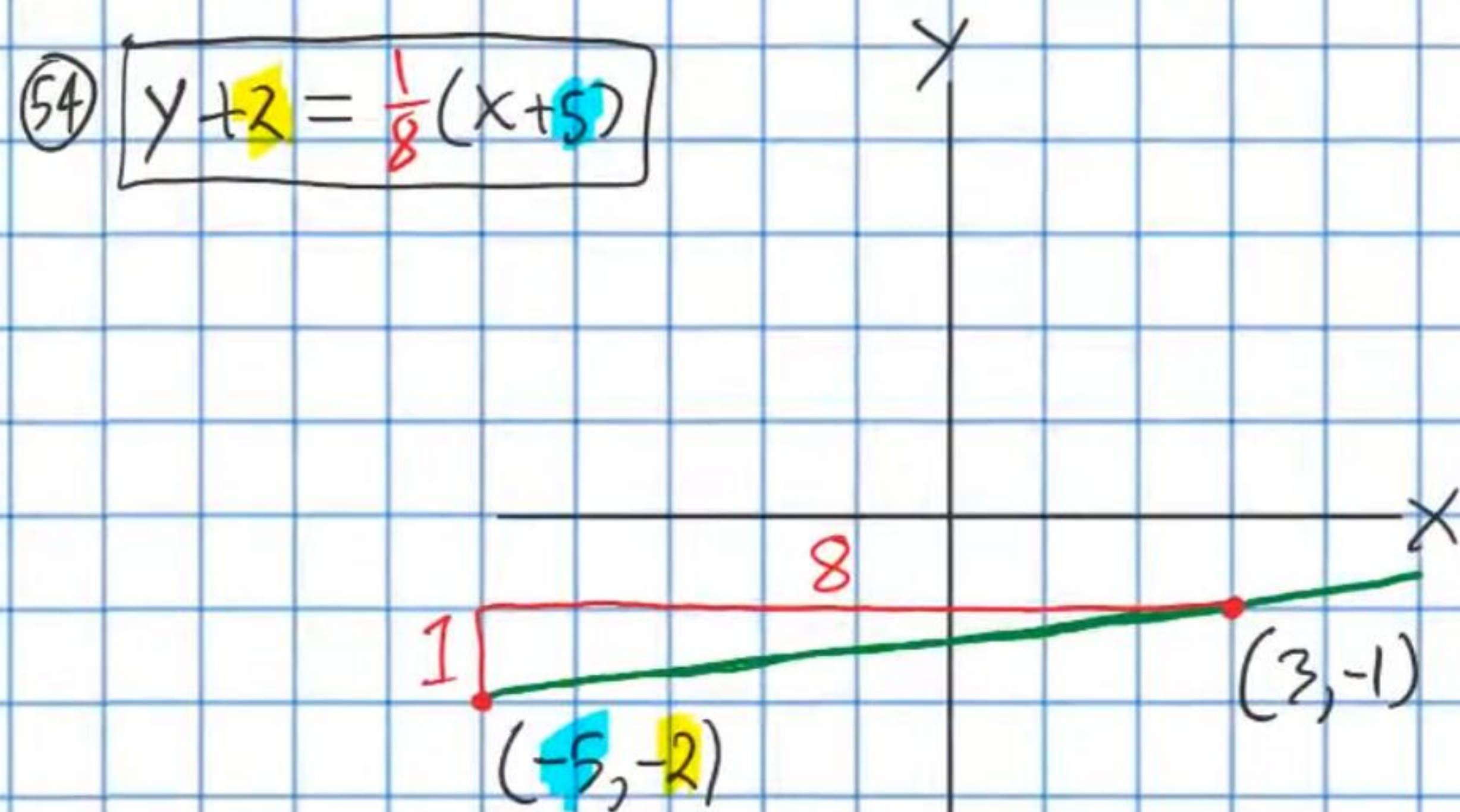
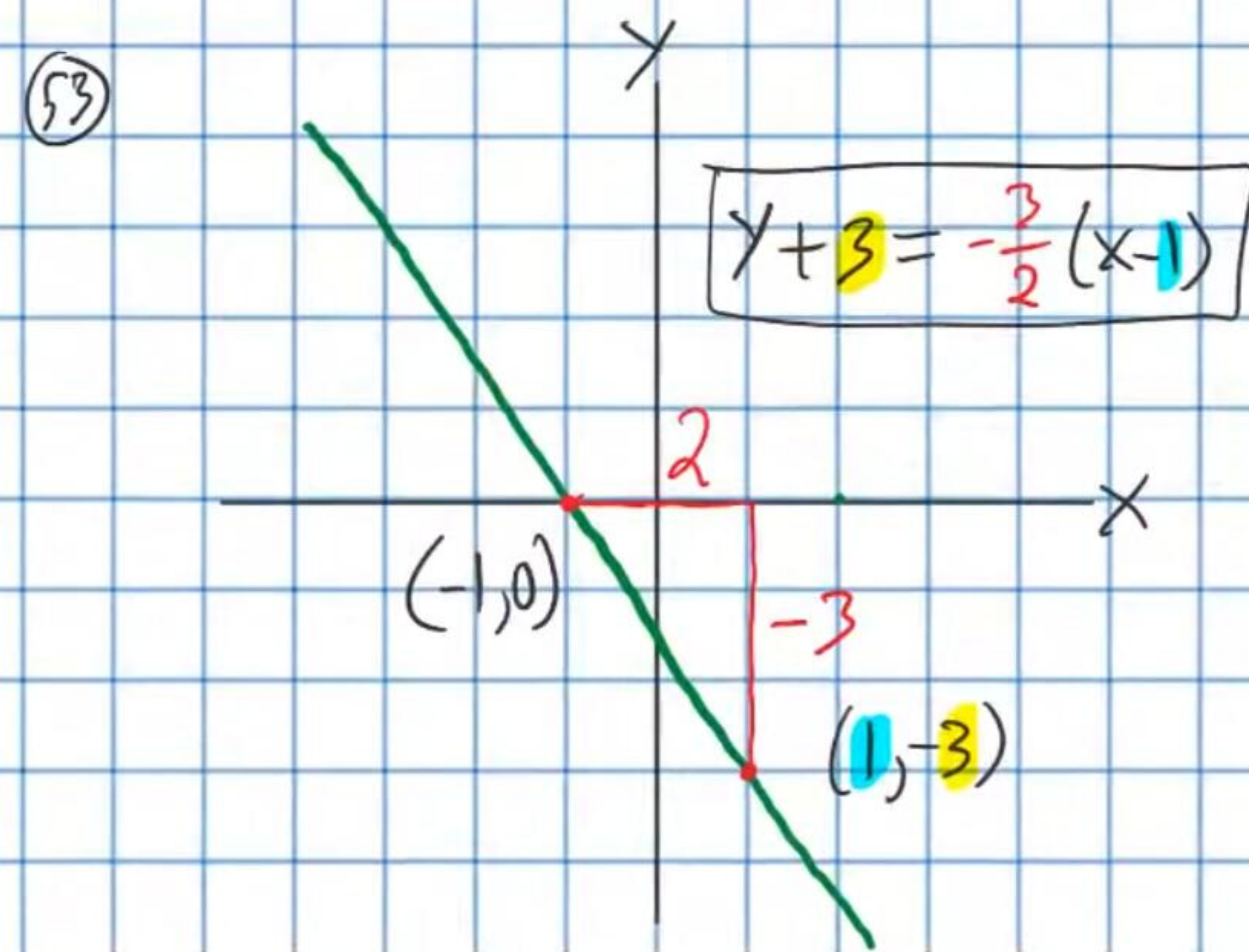
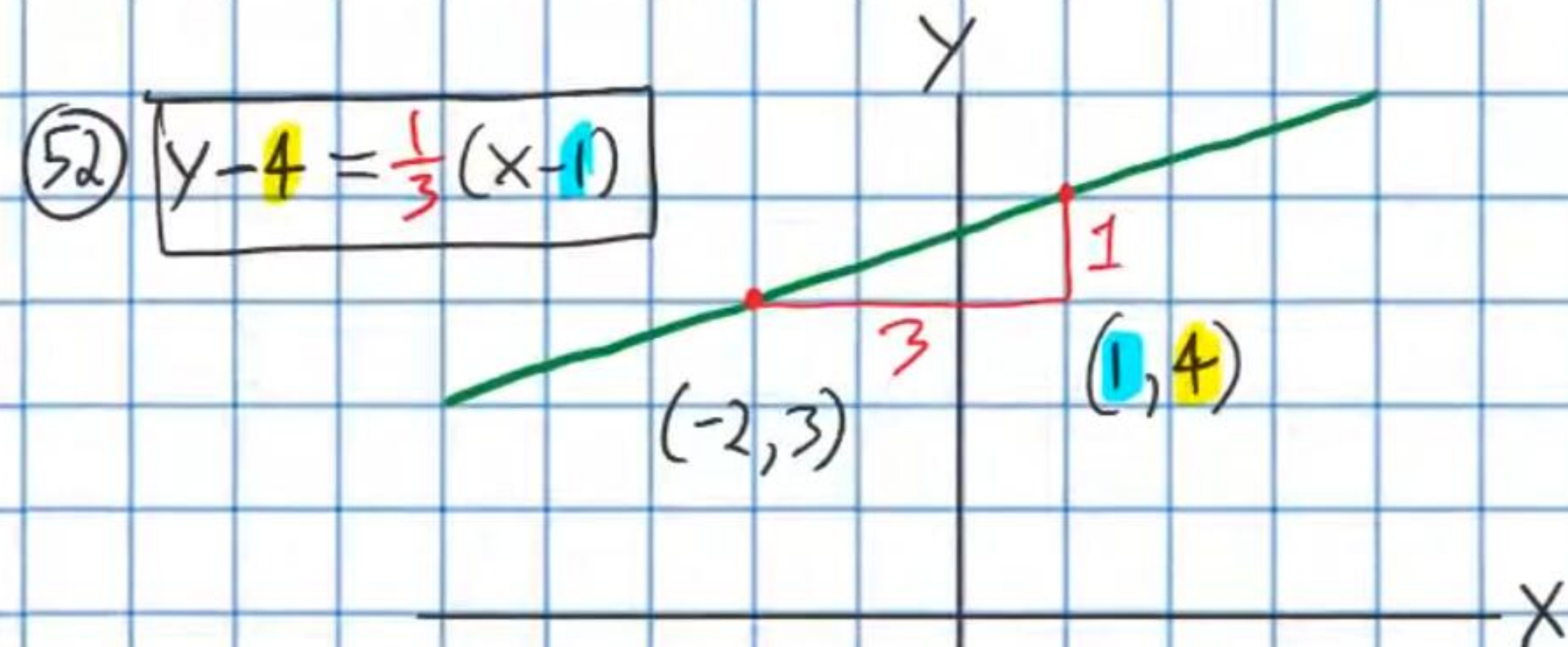
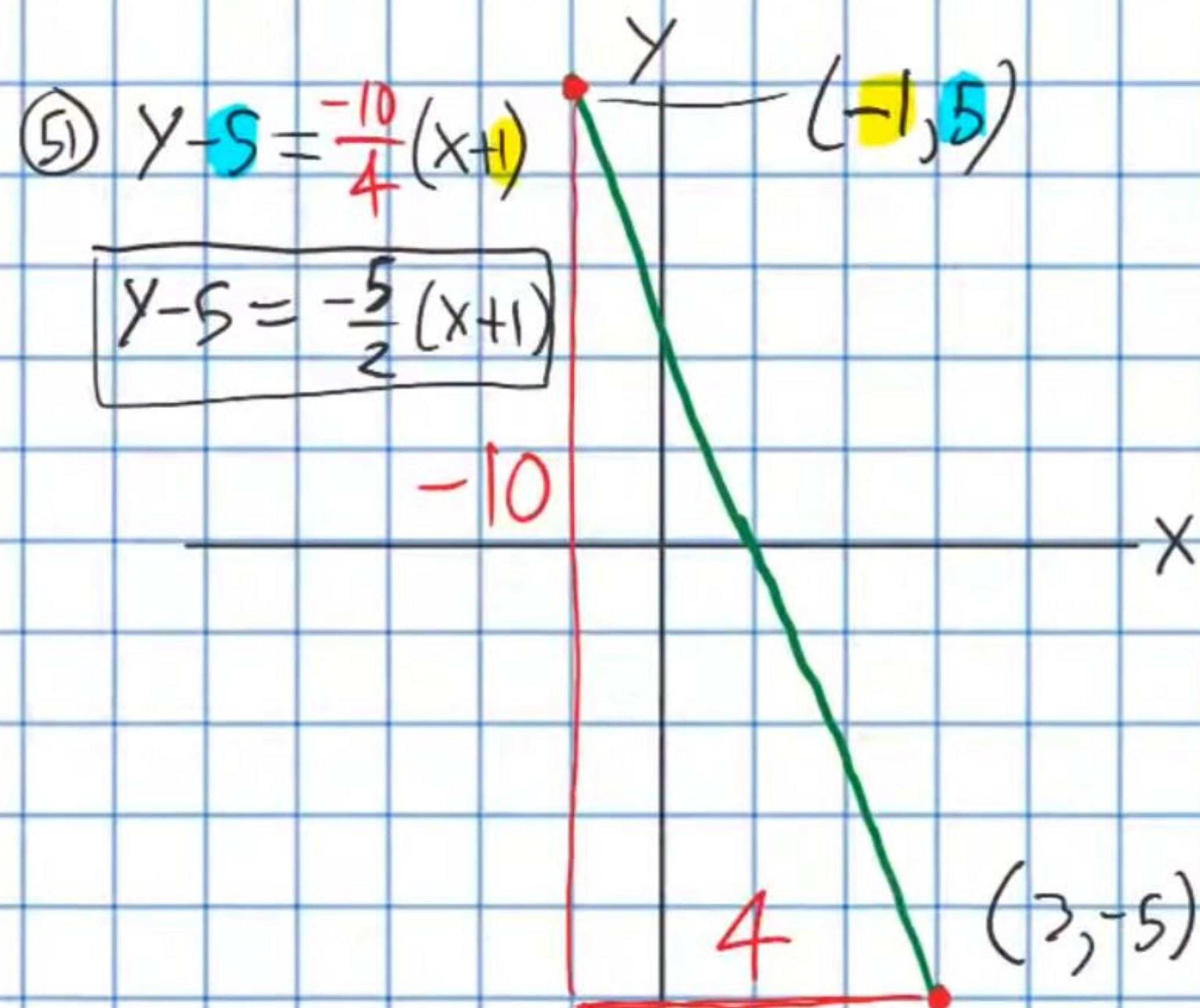


$$(49) \quad y - 3 = \frac{2}{5}(x - 1)$$



$$(50) \quad y + 1 = -\frac{1}{5}(x + 4)$$





For each graph below, write an equation of the line shown and identify its slope.

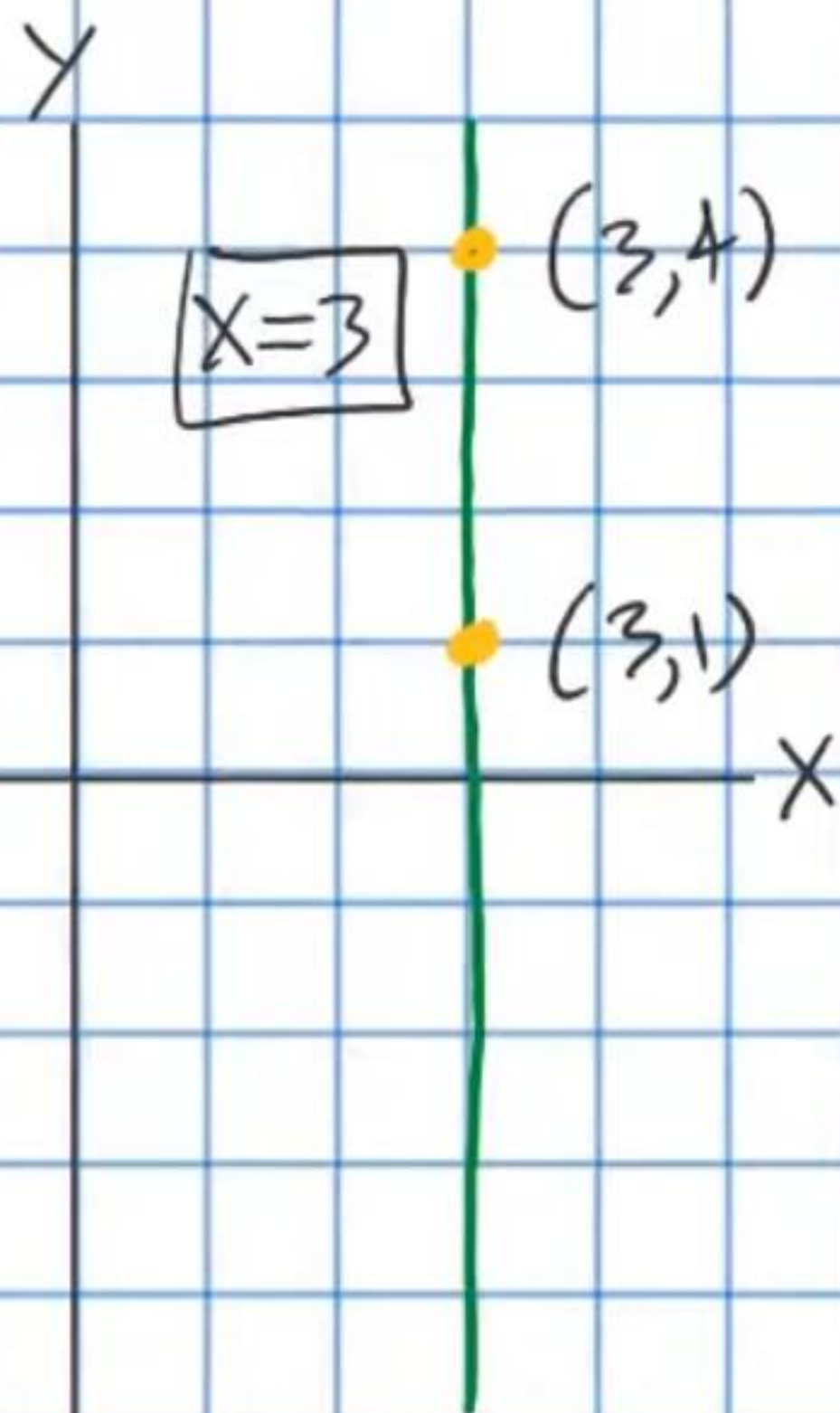
(55)

$$m = \frac{y_2 - y_1}{x_2 - x_1}$$

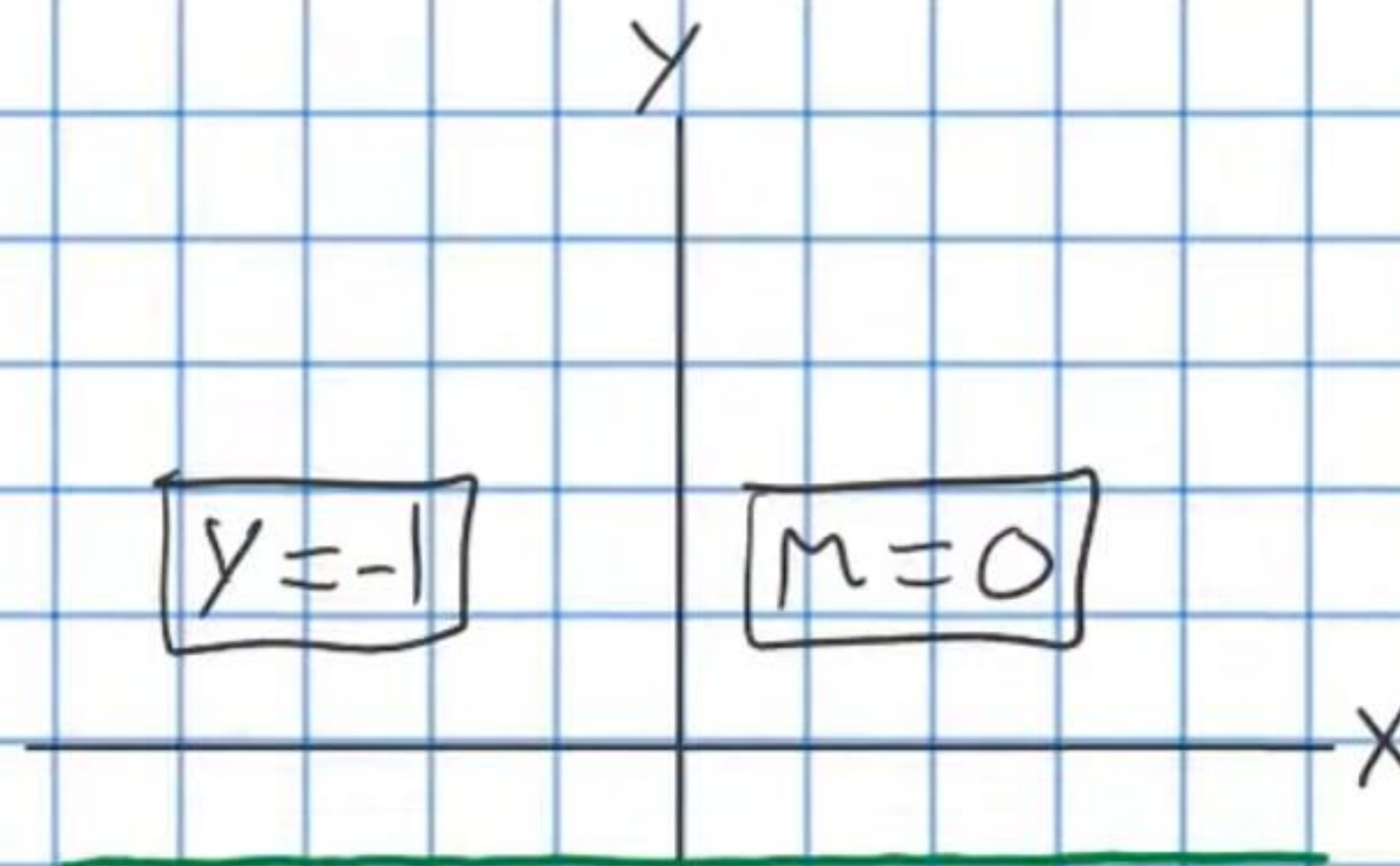
$$m = \frac{4 - 1}{3 - 3}$$

$$m = \frac{3}{0}$$

$$m = \text{Und}$$



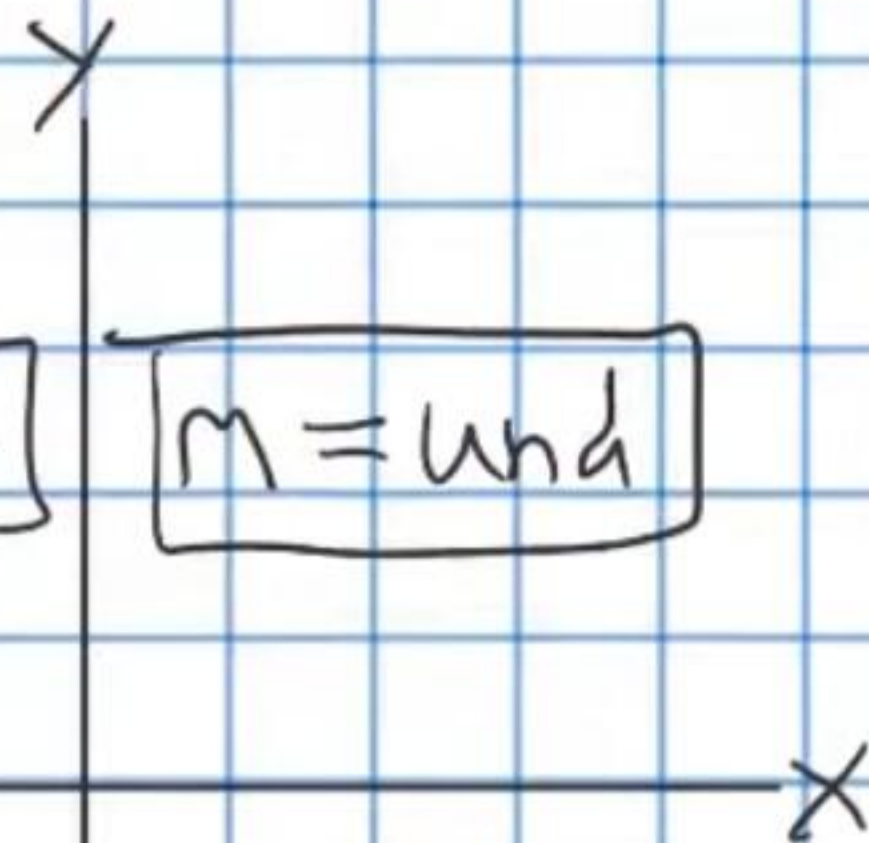
(56)



(57)

$$x = -4$$

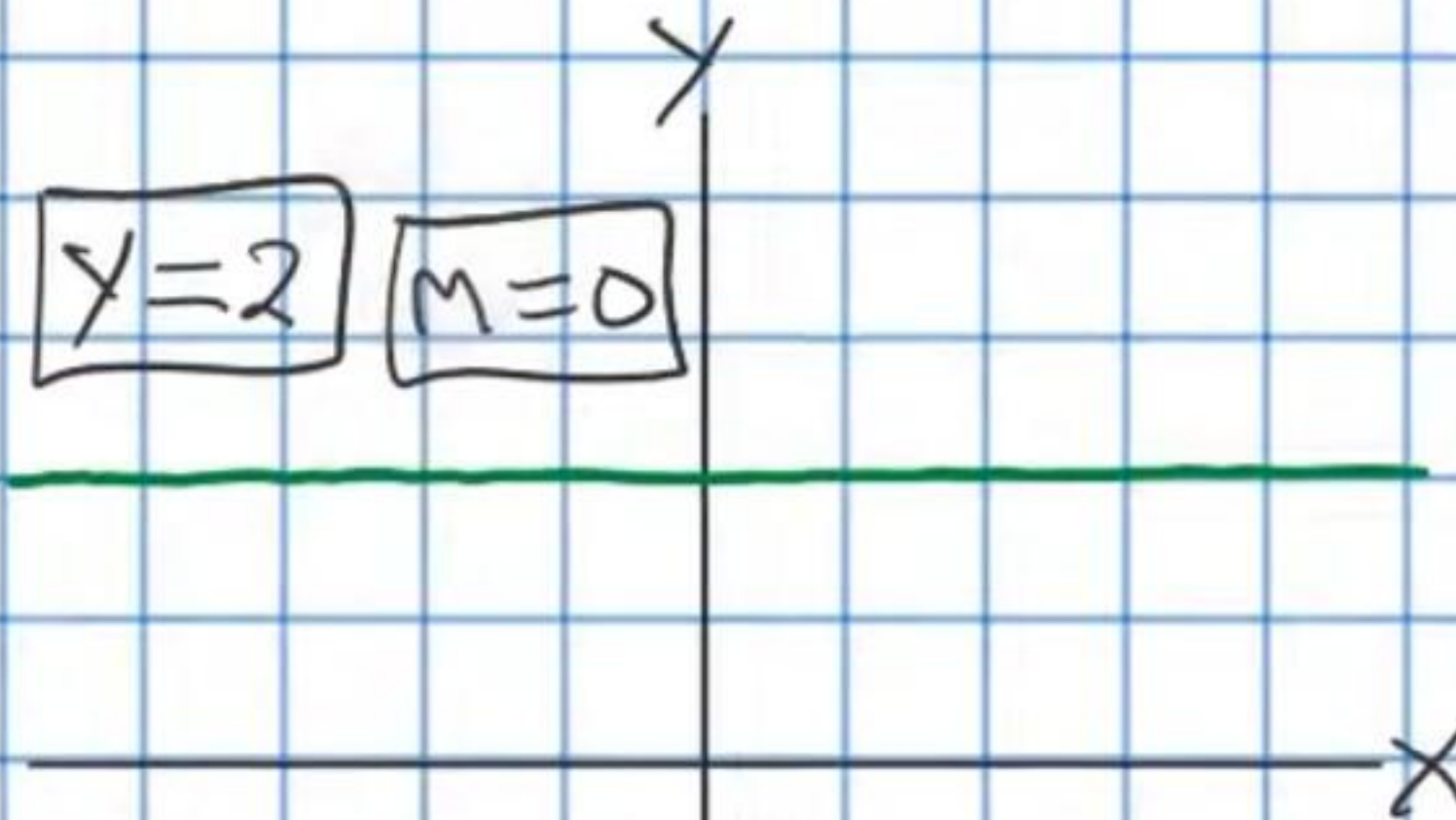
$$m = \text{Und}$$



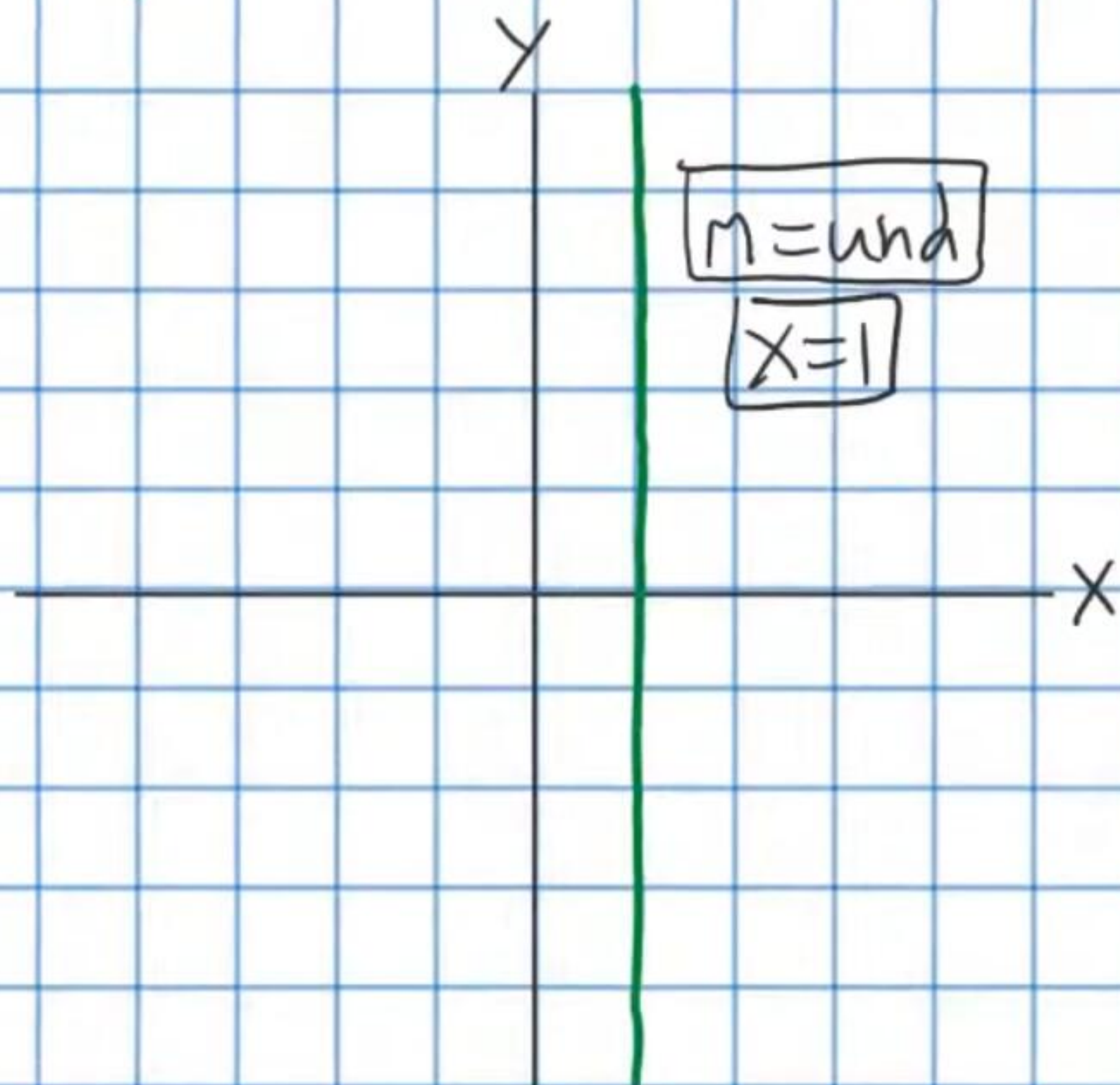
(58)

$$y = 2$$

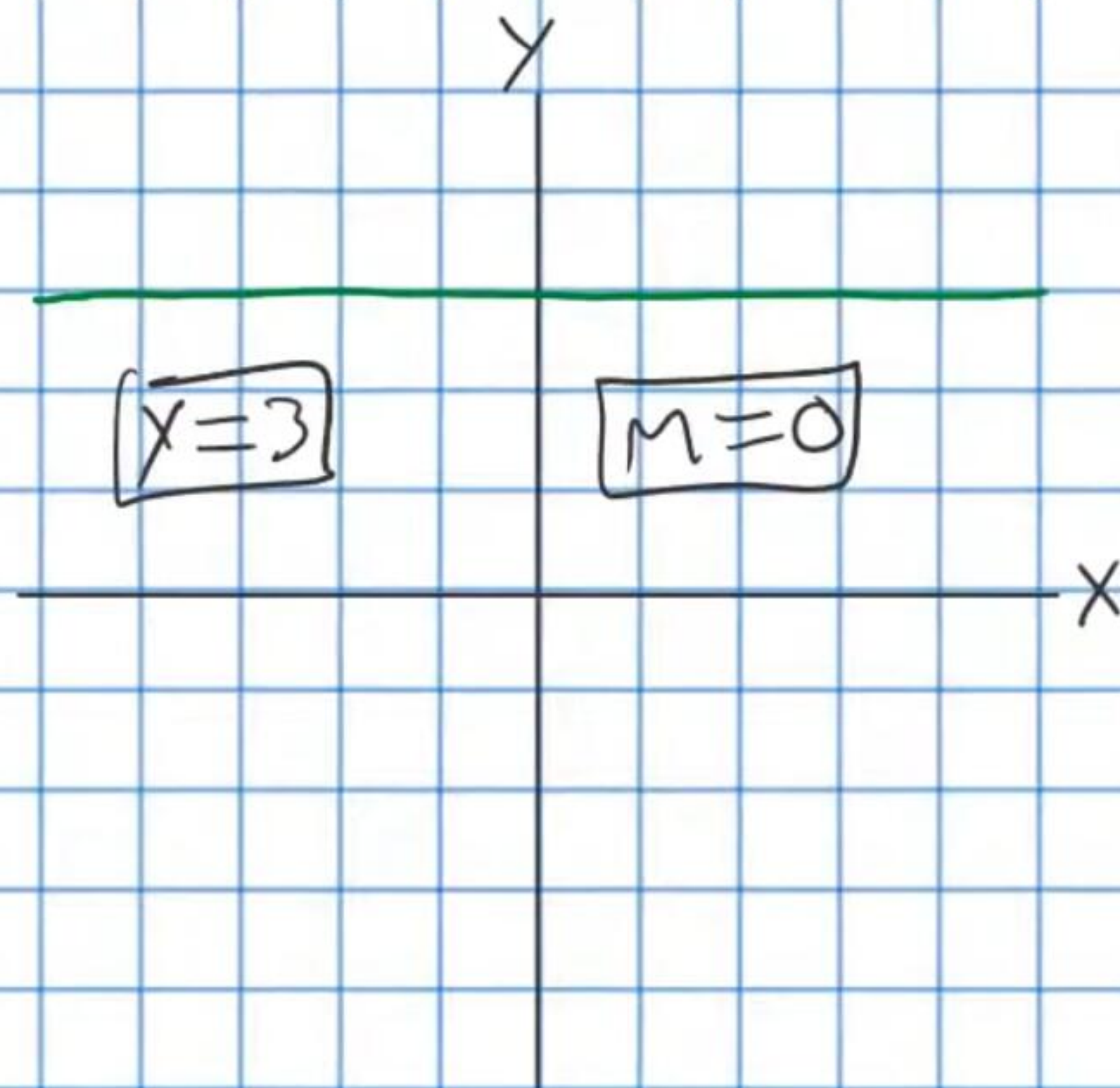
$$m = 0$$



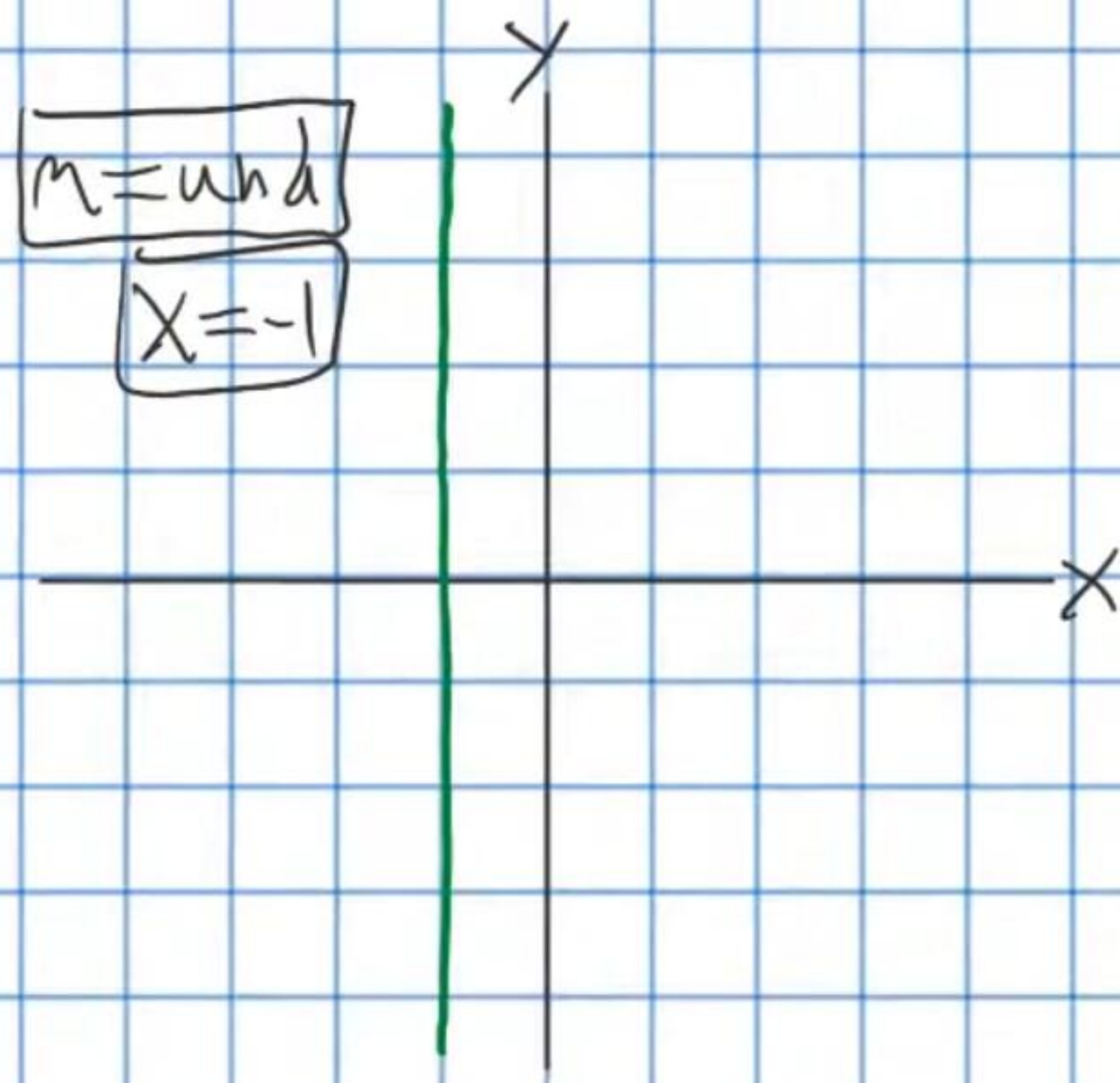
59



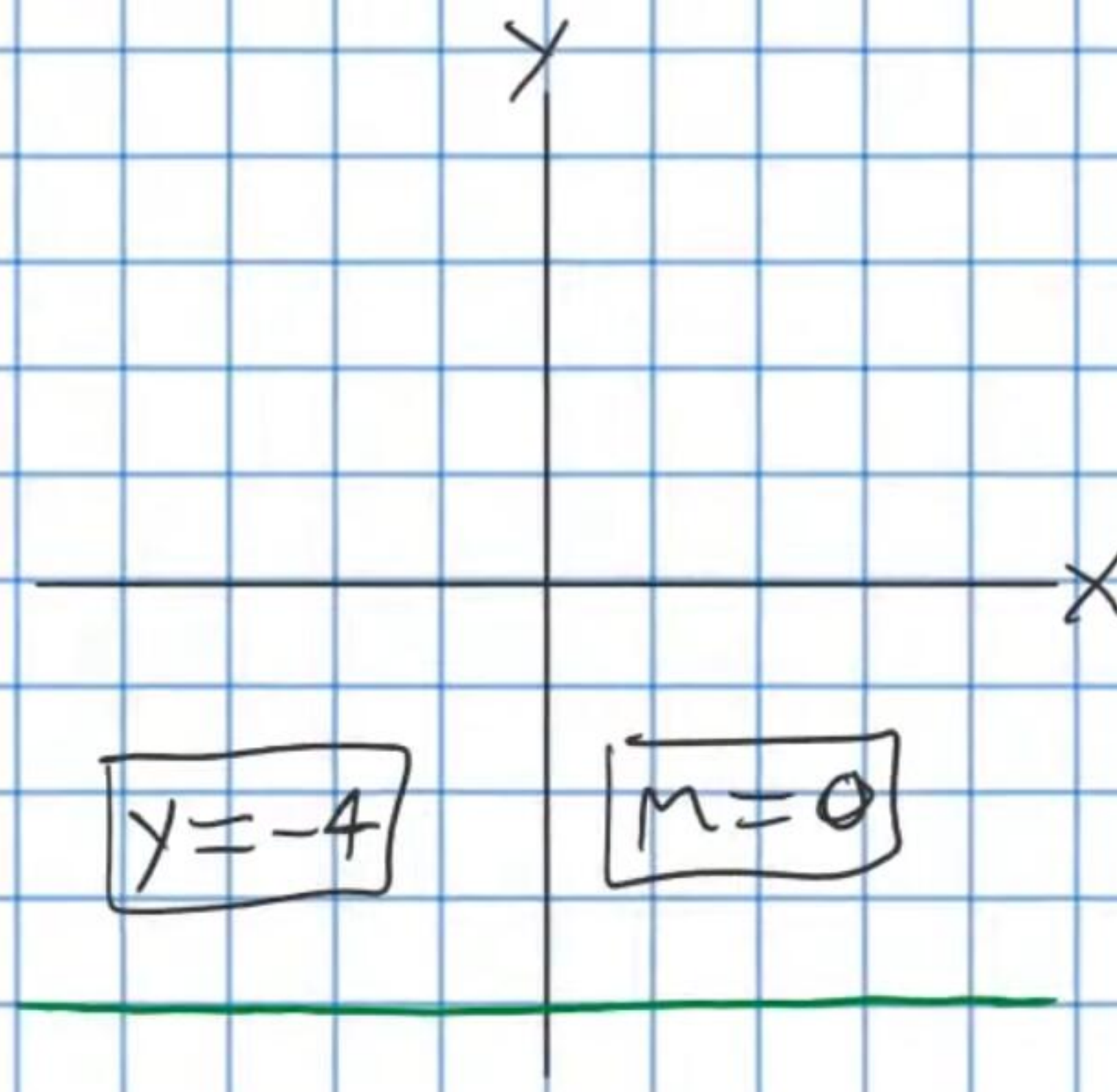
60



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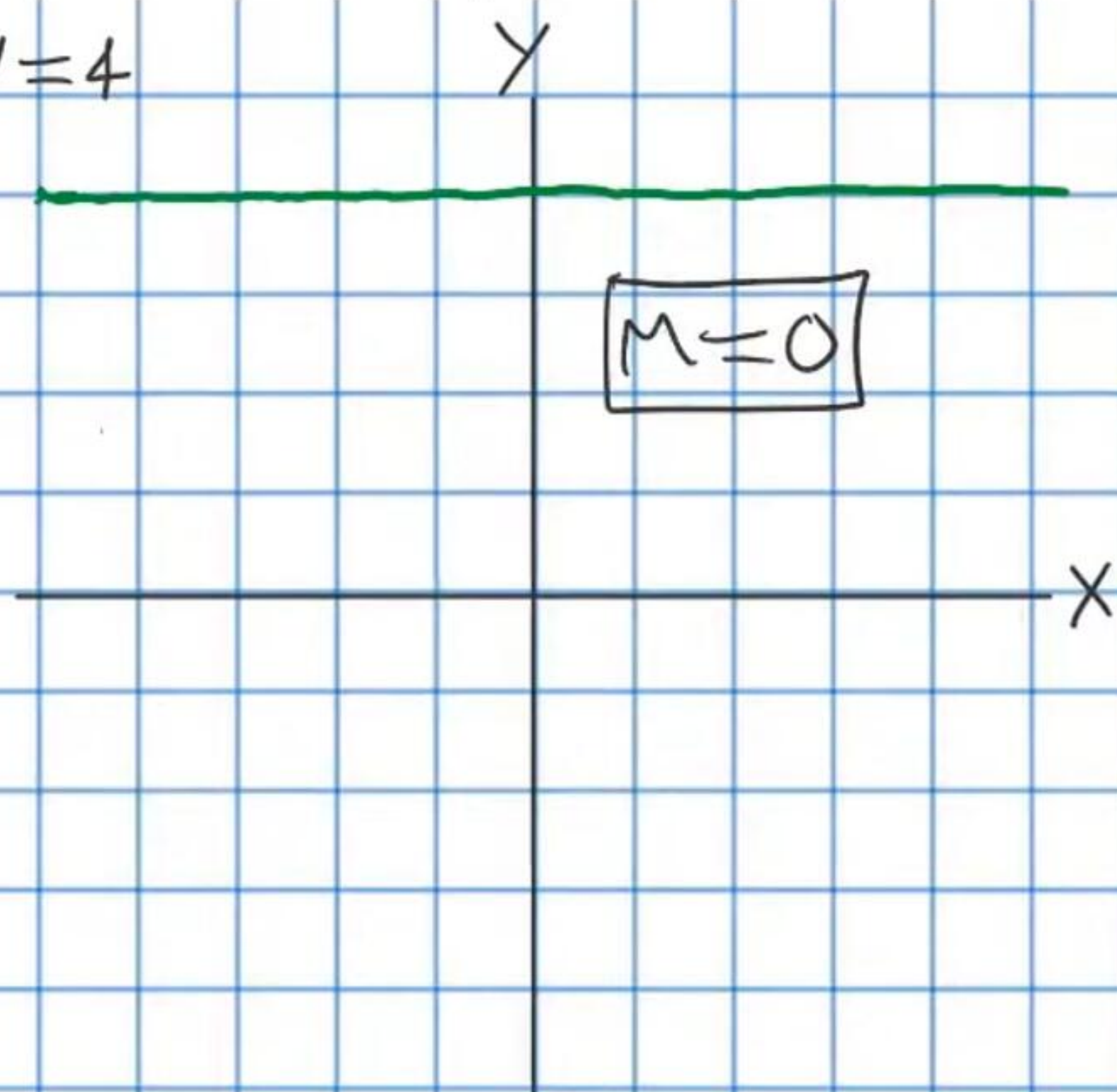


62

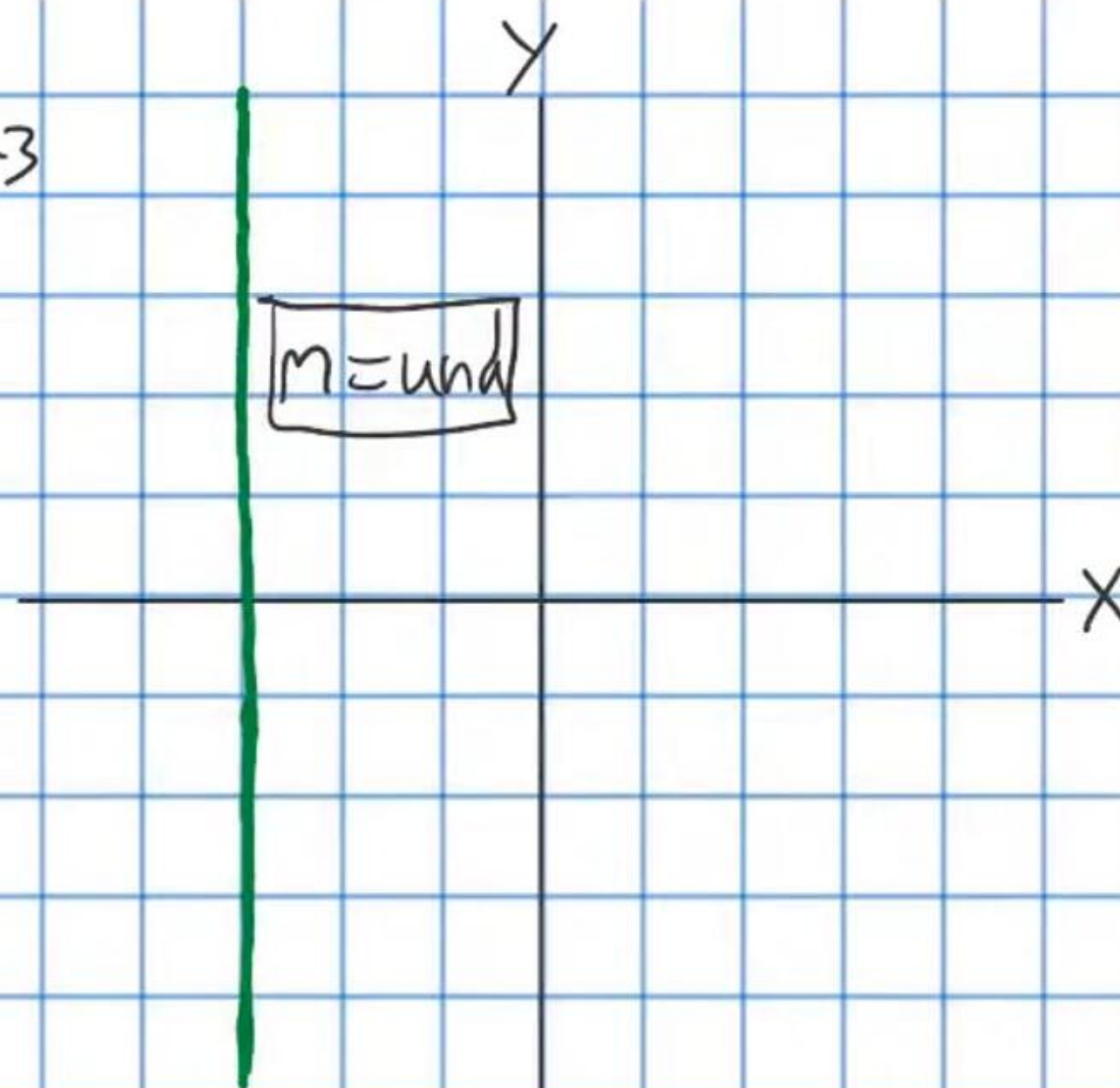


Graph each equation below and identify the slope of the line produced.

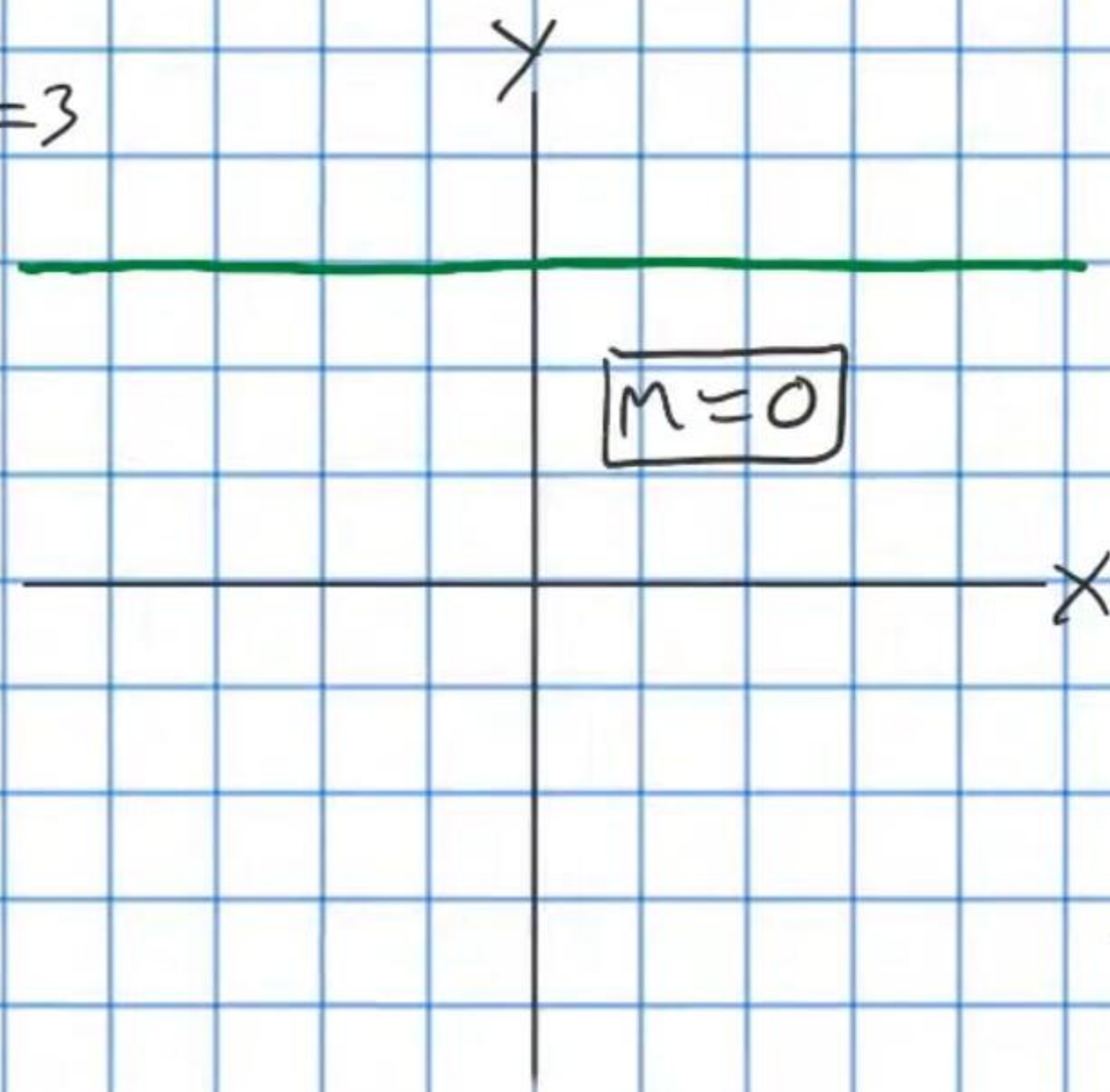
63) $y=4$



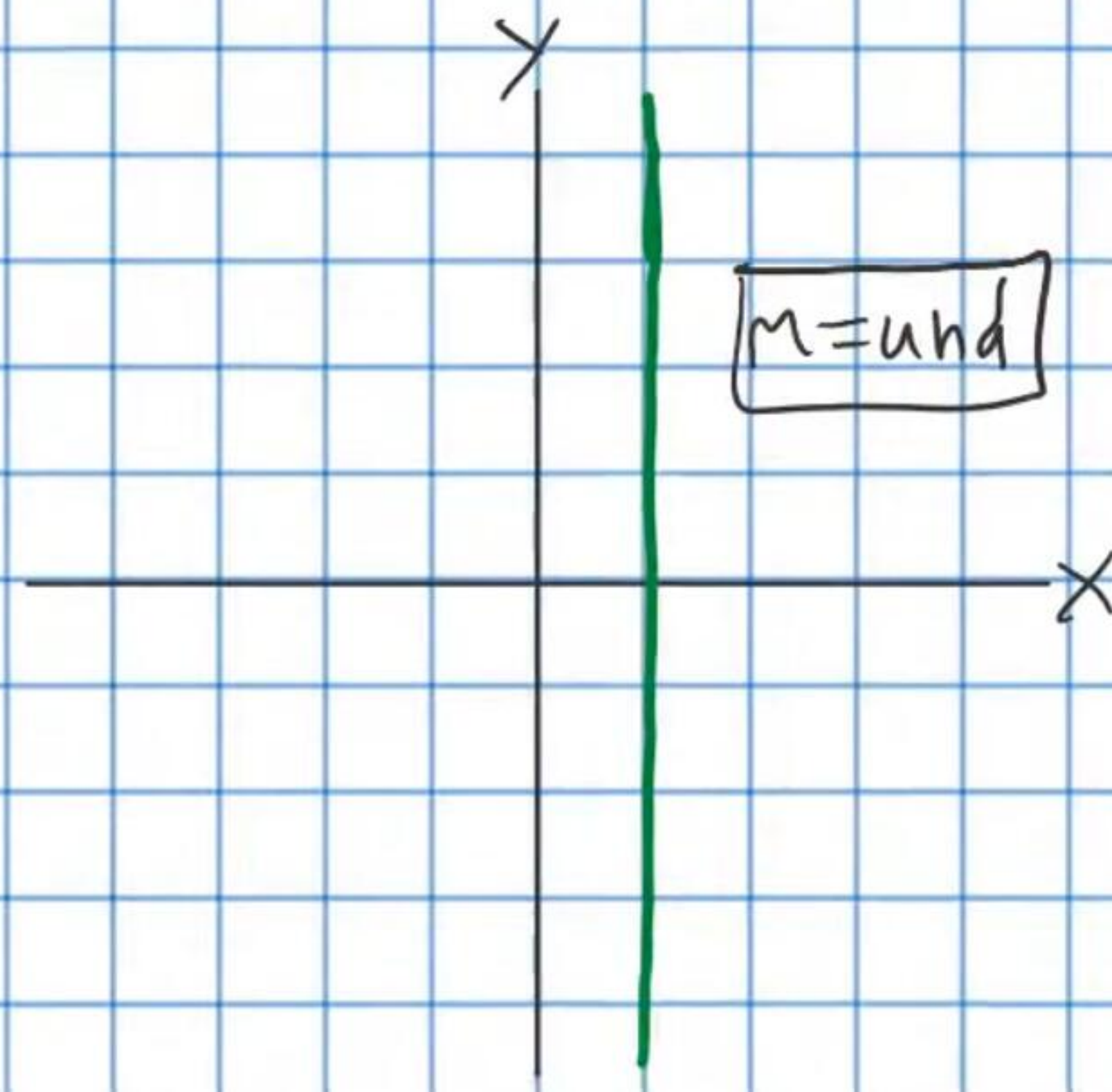
64) $x=-3$



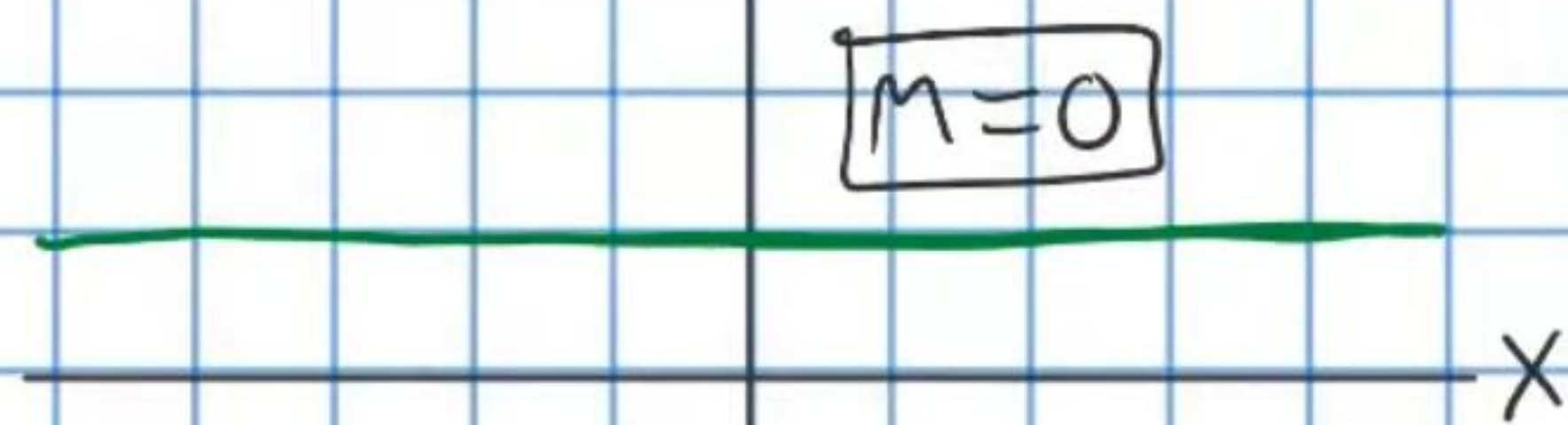
65) $y=3$



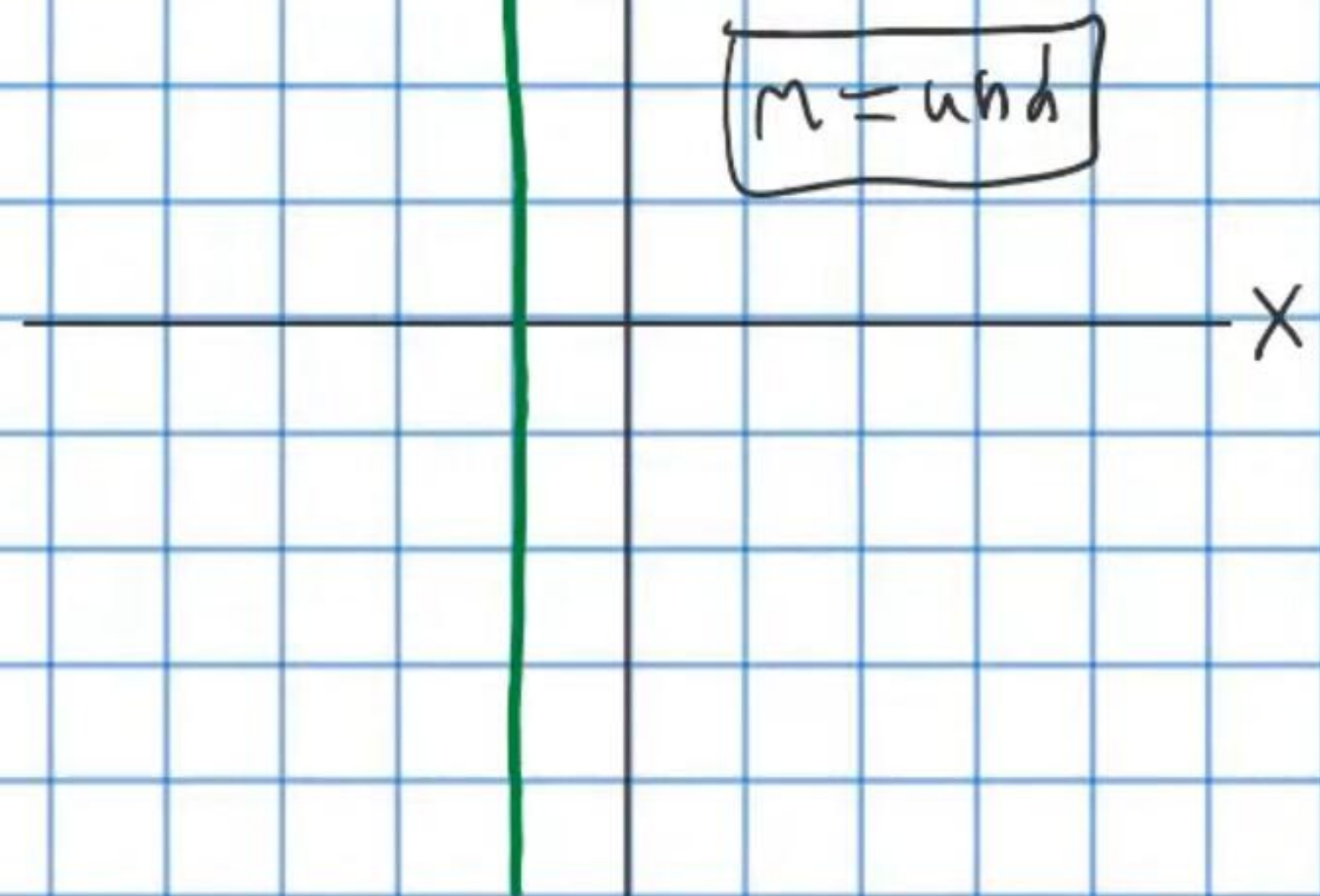
66) $x=1$



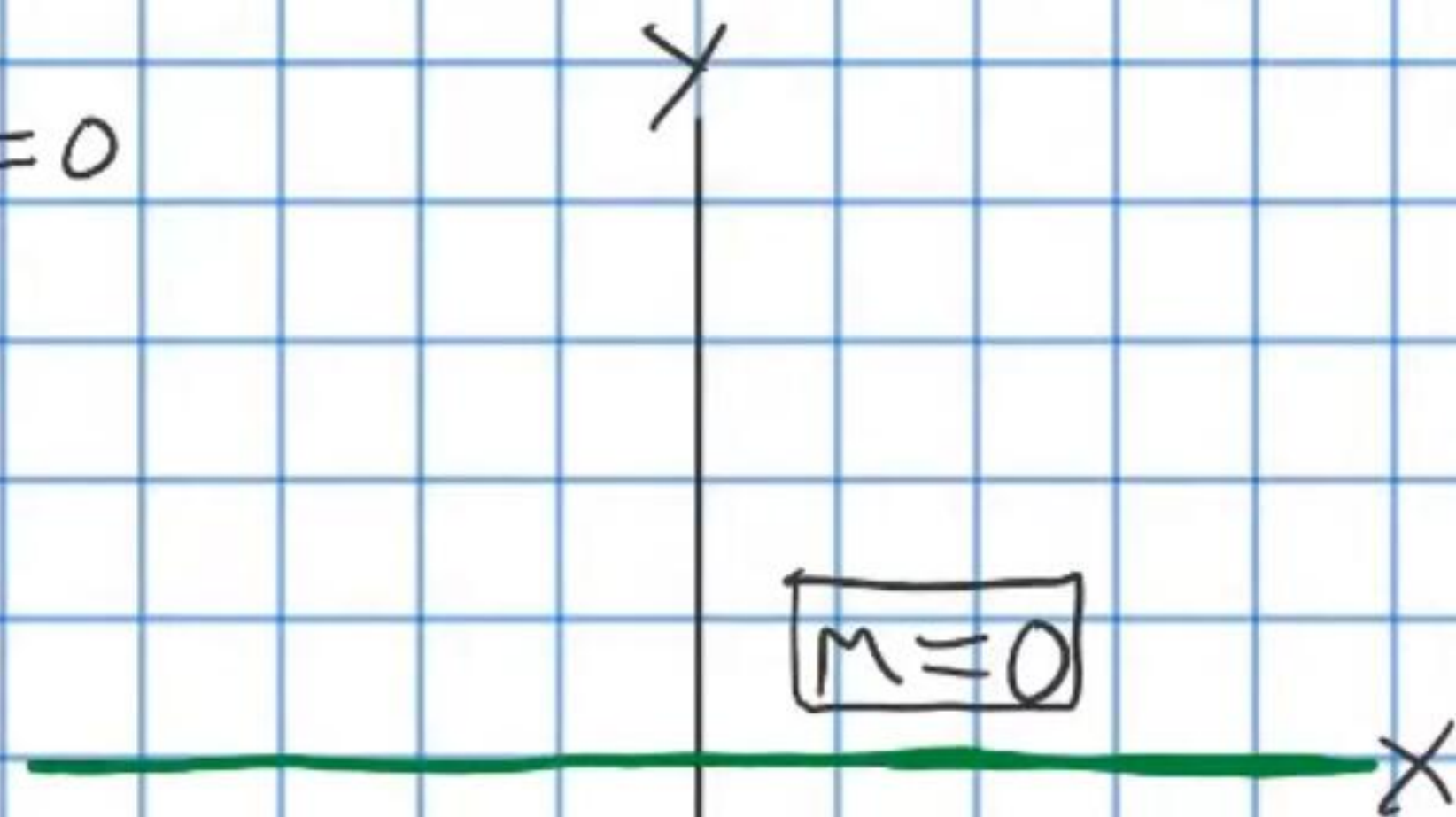
⑥⑦ $y=1$



⑥⑧ $x=-1$



⑥⑨ $y=0$



⑦⑩ $x=2$

